Many-to-many Matching with Cluster-Approximated Spatial Constraints*

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Abstract

We study many-to-many matching with exact agent demands, exact slot quotas, and spatial composition constraints represented by a partition of agents into proximity clusters. At most one agent from each cluster may be assigned to a given slot. Although this local constraint is matroidal and induces a substitutable fixed-priority choice rule, exact demands and quotas couple completion globally across slots. Consequently, Deferred Acceptance and myopic Serial Dictatorship may terminate without a complete allocation even when one exists. In the quota-two benchmark, Deferred Acceptance has a uniform complete-feasibility guarantee if and only if there are exactly two clusters or the spatial constraint is nonbinding because every cluster is a singleton. We characterize completable partial allocations through cluster-based Hall inequalities and implement the resulting completion test using maximum flow. Building on this characterization, we introduce a completion-preserving priority mechanism that, under heterogeneous demands and capacities, produces a feasible and Pareto-efficient allocation satisfying no justified envy and pairwise stability. Simulations calibrated with administrative data on pharmacy duty schedules in Türkiye show that the mechanism clears the market in every reported scenario and improves model-implied rank outcomes under the generated preference profiles. These conclusions are robust to alternative cluster constructions, geographic specifications, and preference calibrations.

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1 Introduction

We study a shift-scheduling problem with fixed geographic service sites in which each shift must select a sufficiently dispersed set of locations to ensure equitable and convenient access. We focus on a tractable class of such problems in which spatial distribution requirements are approximated by proximity clusters. Agents within the same cluster are treated as too close to be assigned jointly, whereas agents from different clusters are treated as mutually compatible. We formalize this requirement as a zone-coverage constraint that restricts which agents may be assigned jointly. Combined with hard quotas on both sides of the market—each agent must work an exact number of shifts and each shift has fixed capacity—these spatial constraints severely limit the set of feasible allocations. In this environment, feasibility is not a peripheral technicality but the core design challenge: standard Deferred Acceptance (DA) and myopic Serial Dictatorship (SD) mechanisms can break down, leading to deadlock even when feasible completions exist, and conventional notions of efficiency, fairness, and stability become mutually incompatible with the geographic structure. Our central objective is therefore to characterize the feasible set in a way that enables implementable mechanisms and a coherent reformulation of the main axioms.

A scheduling problem under cluster-approximated spatial distribution constraints. Our analysis is motivated by a broad class of scheduling problems (see Section 5), most notably the assignment of pharmacy duty shifts in Türkiye, which serves as a running example throughout the paper. Duty shifts are determined annually at the district level, with heterogeneous capacities across days and holidays. Each pharmacy is entitled to a fixed number of shifts per year, and shift-level revenue coefficients provide an observable proxy for the relative attractiveness of duty shifts. To balance cumulative revenues, the regulating authority tracks realized earnings over multiple years and implicitly lowers the priority of pharmacies with higher past revenues. Spatial distribution is enforced through fixed zone-covering pairs—an institutional simplification of a more general separation constraint: pharmacies within a designated pair are typically assigned jointly to the same shifts throughout the year.

While institutionally functional, this centralized system exhibits three structural weaknesses. First,

it disregards pharmacies' heterogeneous preferences over shifts, effectively assuming that revenue fully captures welfare. Second, by fixing zone-coverage to permanent pairs, it imposes rigid spatial constraints that block otherwise feasible reallocations. Third, assignments rely on an *ad hoc* revenue-equalizing heuristic lacking formal justification or transparency. These features collectively limit welfare and obscure the normative basis of allocation decisions. Consistent with the view that market design should remedy institutional flaws "[. . .] while adhering as closely as possible to current policy goals and the existing setup" (Sönmez, 2025), our objective is to provide a mechanism that preserves hard quotas and spatial requirements while restoring formal guarantees of feasibility, fairness, and efficiency.

Model overview and structural insight. We formulate a two-sided, many-to-many matching model with hard quotas and cluster-approximated spatial compatibility constraints. Agents (pharmacies) have preferences over slots (duty shifts), and all slots use a common priority ordering over agents. Each agent must be assigned exactly a fixed number of slots, and each slot has a fixed capacity.[1] Spatial constraints restrict which subsets of agents may be jointly assigned to a slot.

We model spatial requirements as a binary zone-coverage relation over agents. Under the maintained negative-transitivity restriction, this relation induces a partition of agents into proximity clusters (Lemma 1). Agents within the same cluster are too geographically close to be assigned jointly, whereas agents from distinct clusters form admissible pairs. This partition structure is the key to understanding feasibility. When geographic constraints involve overlapping neighborhoods or slot-specific admissibility, the partition representation no longer applies, and the exact characterization results of this paper need not hold in the same form.

The induced geometry yields an exact characterization of DA's uniform feasibility guarantee. In the single-unit-demand environment with homogeneous slot quota two, DA produces a feasible allocation for every common priority ordering and preference profile if and only if either the zone-coverage relation induces exactly two clusters or every cluster is a singleton (Theorem 1). The latter is a non-binding case: when all clusters are singletons, every pair of distinct agents is compatible and the spatial

[1] In our application, each pharmacy is guaranteed a fixed number of duty shifts within a calendar year; accordingly, institutional practice treats these requirements as hard quotas.

constraint imposes no effective restriction.

Conditional on a non-vacuous zone-coverage constraint, the two-cluster case is therefore the exact boundary for DA's uniform feasibility guarantee. With two clusters and quota two, every full feasible slot contains one agent from each cluster, so completing a slot preserves the cluster balance required by the residual problem. When the constraint is non-vacuous and there are three or more clusters, a locally admissible slot may be filled using only a subset of clusters. The resulting exclusion may leave residual cluster demands and residual slot capacities that cannot be jointly completed. Thus, the relevant distinction is between local admissibility and global completion: the former need not preserve the latter.

Mechanism design under global feasibility. To address this structural failure, we introduce Feasible Serial Dictatorship (FSD) for the single-unit-demand case, a mechanism that assigns agents sequentially while enforcing global completion at every step. Whenever a feasible allocation exists, FSD produces one consistent with the priority order (Theorem 3). By embedding a forward-looking feasibility test into each assignment decision, the mechanism eliminates structural dead-ends. In this sense, FSD is a completion-preserving version of serial dictatorship: it accepts a tentative assignment only when the resulting partial allocation remains extendable to a feasible full allocation.

We extend the analysis to multi-unit demand environments. The resulting Generalized Feasible Serial Dictatorship ($GFSD$) ensures feasibility through a generalized completion condition characterized using tools from network flow theory, specifically the Ford–Fulkerson Theorem and the Integrality Theorem (Schrijver, 2003). This characterization (Theorem 4) translates the geometric constraint into a global flow condition that is both necessary and sufficient for implementability. $GFSD$ uses this condition dynamically: each tentative assignment is accepted only if the residual problem still admits a feasible completion.

Normative properties. The allocations produced by FSD and $GFSD$ satisfy framework-appropriate versions of no justified envy, Pareto efficiency, and pairwise stability under spatial constraints (Theorems 3 and 5). We also clarify the mechanisms' incentive properties: FSD is strategy-proof in the single-unit environment, whereas $GFSD$ is generally manipulable under multi-unit demand. Where

necessary for the normative analysis, we extend preferences over slots to bundles using a parsimonious ordinal lexicographic rule that requires neither cardinal utilities nor additional assumptions about complementarities across shifts.

Accordingly, the empirical analysis is interpreted as a counterfactual comparison under preferences generated by the maintained Data Generated Process (DGP), rather than as an equilibrium prediction under strategic reporting. These results show that strong normative properties can be restored without relaxing the institutional constraints, while also identifying the incentive limits of sequential completion-preserving allocation in the multi-unit domain.

Summary of contributions. This paper makes three main contributions. First, we introduce a matching model with hard quotas and spatial incompatibility constraints and provide a structural characterization of feasibility: under the maintained negative-transitivity restriction, the zone-coverage relation induces proximity clusters that determine exactly when allocations can be completed. Second, we identify a sharp threshold for DA—it has a uniform feasibility guarantee if and only if either there are exactly two clusters or the spatial constraint is nonbinding because every cluster is a singleton—and show that with three or more clusters classical local acceptance logic can generate inherent dead-ends. This isolates a precise boundary at which standard local acceptance logic no longer preserves feasible completion. We further show in the appendix that this threshold is the quota-two instance of a broader capacity–cluster balance. Third, we characterize the conditions under which partial allocations admit global completion and design a mechanism that ensures global completability at every step. We show that this mechanism restores feasibility while preserving the normative properties described above. Together, these results provide a tractable framework for mechanism design in environments where geographic structure fundamentally constrains implementability.

Empirical validation. We conclude by simulating the pharmacy scheduling environment using administrative data from Türkiye. Spatial constraints are represented by partitions recovered from historical co-duty assignments and geographic compactness. Preferences are modeled using revenue-based attractiveness with noise, and priorities reflect historical revenue equalization. We compare the current heuristic mechanism with algorithmic benchmarks. Across the reconstructed environments, rigid part-

ner rules and myopic allocation procedures may fail to clear the market or produce poor model-implied rank outcomes. $GFSD$ clears the market in every reported scenario and improves model-implied rank outcomes under the simulated preference profiles. The qualitative feasibility ranking is stable across the threshold, geographic, and preference specifications considered.

Relation to the literature. We contribute to market design by studying priority-based allocation under rich feasibility constraints with a spatial structure. Our work is related to the canonical literature on matching with contracts, many-to-many stability, and constrained allocation. Matching with contracts provides a general framework in which substitutability and related conditions support stable outcomes and incentive properties (Hatfield and Milgrom, 2005). Many-to-many matching raises additional stability issues, and substitutability-type restrictions play a central role in ensuring the existence and structure of stable outcomes (Echenique and Oviedo, 2006). A significant strand of the literature examines whether classical rules—Deferred Acceptance (DA) and Serial Dictatorship (SD)—remain well-behaved under distributional or capacity constraints. The dominant paradigm identifies substitutability-type conditions (Hatfield and Kojima, 2010)—often formalized through hierarchical (laminar) caps (Kamada and Kojima, 2018), matroidal structures (Kojima, Tamura, and Yokoo, 2018), or polyhedral quotas (Kojima, Sun, and Yu, 2020)—under which decentralized acceptance procedures retain desirable stability, incentive, or feasibility properties.

The appropriate comparison with this literature requires distinguishing local feasibility from global completion. The connection to matroidal and laminar structures is exact at the local level. At an individual slot, the proximity clusters are pairwise-disjoint subsets of the common ground set of agents. The restriction that at most one agent may be selected from each cluster defines a partition matroid, truncated by the slot's overall capacity. Equivalently, the locally admissible agent sets form a laminar matroid (Barman and Biswas, 2020; Fife and Oxley, 2017). The associated fixed-priority greedy choice is therefore substitutable (Yokoi, 2019); in the quota-two benchmark, Proposition 1 establishes this property directly. Our environment therefore does not depart from the matching literature through a failure of local matroidality or local substitutability.

The additional issue is complete feasibility under exact requirements on both sides. Local substi-

tutability disciplines acceptance decisions at an individual slot, but it does not imply that independently admissible slot holdings can be extended to an allocation satisfying every agent's exact demand and every slot's exact quota. Assigning an agent to a slot uses not only ordinary slot capacity but also the cluster-specific opportunity at that slot: once a cluster is represented, no additional agent from that cluster can be assigned there. Consequently, a collection of locally admissible slot holdings may leave residual cluster demands and residual capacities that cannot be jointly completed. The relevant object is therefore not merely a collection of independent local choices, but the set of partial allocations that admit a feasible completion. This is why DA may terminate without a complete feasible allocation even though every slot uses a substitutable choice function and a complete feasible allocation exists.

This global completion problem nevertheless has a precise matroidal structure. For each partial allocation, each slot induces a residual laminar matroid describing which remaining units of demand can still be assigned there. A completion exists exactly when all remaining demand units can be distributed across these slot-specific matroids; formally, this is a matroid-union problem. Their union is itself a matroid, but it need not be laminar and changes with the partial allocation because each tentative assignment alters both residual capacity and cluster availability. Edmonds's matroid-union theorem provides a rank characterization of completion (Edmonds, 1968). Appendix C shows that these rank inequalities reduce exactly to the cluster-based Hall condition in Theorem 4, thereby providing a matroidal interpretation of the maximum-flow characterization underlying $GFSD$.

Our results are therefore complementary to the substitutability and matroidal-constraint approaches. The novelty lies not in introducing a non-substitutable local choice environment, but in characterizing and dynamically preserving exact global completion. Theorem 1 identifies the capacity–cluster boundary at which DA preserves completion, while Theorem 4 characterizes the partial allocations that remain completable. FSD and $GFSD$ differ from myopic local procedures by enforcing the corresponding completion conditions dynamically, using a polynomial-time completion oracle based on maximum flow in the general environment.

Our paper is also related to the literature on distributional constraints, controlled choice, and hard quotas. In regional matching and controlled school choice, constraints typically regulate the number or

type distribution of agents assigned across regions, schools, or institutions (Kamada and Kojima, 2015; Echenique and Yenmez, 2015). Hard distributional constraints and dynamic quota mechanisms have been studied as a way to improve upon artificial-cap approaches while preserving desirable fairness and incentive properties (Fragiadakis and Troyan, 2017). The constraint studied here has a different operational structure: it is per-slot and compositional, so that no slot may contain two agents from the same cluster. Once this per-slot composition rule is combined with hard agent demands and fixed slot capacities, feasibility becomes globally coupled across the entire schedule. Relatedly, daycare allocation models impose ratio constraints through global caps (Kamada and Kojima, 2024). Unlike adjustable cutoffs in those settings, our framework maintains fixed priorities and preserves feasible completion through a completion characterization.

Closely related work studies serial allocation under general feasibility constraints. Imamura and Kawase (2024), for example, formulate constrained serial dictatorship on a fixed family of feasible matchings, while Imamura and Kawase (2025) study efficiency and strategy-proofness under general constraints and relate their results to constrained serial dictatorship. $GFSD$ uses the same broad constrained serial allocation logic but supplies a domain-specific completion oracle. In such settings, myopic greedy assignments can generate structural dead-ends, requiring feasibility to be enforced dynamically through a global completion condition.[2]

Multi-unit demand introduces further complications, as sequential priority assignment may conflict with bundled preferences (Manea, 2007). We address this by reducing multi-unit requests to single-unit choices through a replica-market construction while maintaining priority consistency, allowing us to focus on global feasibility under spatial constraints.

2 Model

We study a scheduling problem in which a finite set of slots $\mathcal{S}$ are allocated to a finite set of agents $\mathcal{A}$. Each agent $i \in \mathcal{A}$ has demand $d_i \in \mathbb{Z}_+$, and each slot $s \in \mathcal{S}$ has capacity $q_s \in \mathbb{Z}_+$, with $\sum_{i\in\mathcal{A}} d_i = \sum_{s\in\mathcal{S}} q_s$. Let $d = (d_i)_{i\in\mathcal{A}}$ and $q = (q_s)_{s\in\mathcal{S}}$ denote the demand and capacity vectors.

[2]Similar breakdowns of myopic allocation logic arise in seat allocation with gender-based restrictions (Afacan, Dur, and Dur, 2025), where adaptive procedures are introduced to preserve feasibility under additional constraints.

Each $i \in \mathcal{A}$ has a strict preference relation π_i over $\mathcal{S}$, with weak and strict parts π_i^R and π_i^P. The preference profile is $\pi = (\pi_i)_{i \in \mathcal{A}}$. A *priority ordering*, applied at every slot, is a linear order $\succ$ on $\mathcal{A}$, with weak part $\succeq$, where $i \succeq j$ if and only if $i \succ j$ or $i = j$.

Two agents form a *zone-covering pair* if they are sufficiently distant to ensure adequate service-area coverage; such agents may therefore be assigned to the same slot. A slot must be covered by a set of agents in which every pair is zone-covering. We assume that zone-coverage is slot-independent: for any $s, s' \in \mathcal{S}$, agents i, j are zone-covering for s if and only if they are zone-covering for s'. This structure is represented by a symmetric and irreflexive binary relation $\mathcal{F} \subseteq \{(i, j) \in \mathcal{A} \times \mathcal{A} : i \neq j\}$, where $(i, j) \in \mathcal{F}$ if and only if i and j form a zone-covering pair. We refer to $\mathcal{F}$ as the *zone-coverage relation*, and call j a *zone-covering partner* of i whenever $(i, j) \in \mathcal{F}$. For any $A \subseteq \mathcal{A}$, let $\mathcal{F}_A = \{(i, j) \in A \times A : (i, j) \in \mathcal{F}\}$ denote the set of zone-covering pairs in A.

A *scheduling problem under spatial distribution constraints* is specified by $(\mathcal{A}, \mathcal{S}, \mathcal{F}, d, q, \pi, \succ)$. An *allocation* is a function $\mu : \mathcal{A} \to \mathcal{P}(\mathcal{S})$ that assigns each agent a set of slots; $\mu(i)$ denotes agent i's assignment. For each slot $s \in \mathcal{S}$, let $\mu_s = \{i \in \mathcal{A} : s \in \mu(i)\}$ denote the set of agents assigned to s. A *mechanism* is an allocation rule that selects an allocation for each reported preference profile, holding fixed the remaining primitives of the problem. Formally, for fixed $(\mathcal{A}, \mathcal{S}, \mathcal{F}, d, q, \succ)$, a mechanism maps each preference profile π to an allocation, denoted $\mu^\varphi(\pi)$. We write $\mu_i^\varphi(\pi)$ for the assignment of agent i under this allocation.

Definition 1 *An allocation μ is **feasible** if,*

1. *For each $i \in \mathcal{A}$, $|\mu(i)| = d_i$ (demand condition)*
2. *for each $s \in \mathcal{S}$, $|\mu_s| = q_s$ (quota condition)*
3. *$i, j \in \mu_s$ implies $(i, j) \in \mathcal{F}$ (zone-coverage condition)*

2.1 Structure of the spatial distribution constraints

We approximate $\mathcal{F}$ by imposing the substantive restriction that whenever i and j are too close to form a zone-covering pair and j and k are too close as well, then i and k are also deemed too close.

Formally, $(i, j) \notin \mathcal{F}$ and $(j, k) \notin \mathcal{F}$ imply $(i, k) \notin \mathcal{F}$, hence $\mathcal{F}$ is *negatively transitive*. Under the natural symmetry of "too close," this is equivalent to a partition of agents into proximity clusters: agents within the same cluster are geographically too close to share a slot (all within-cluster pairs lie in the complement relation of $\mathcal{F}$), whereas agents from different clusters form zone-covering pairs (all cross-cluster pairs lie in $\mathcal{F}$).

Lemma 1 *Let $\mathcal{F}$ be an irreflexive binary relation on $\mathcal{A}$. $\mathcal{F}$ is symmetric and negatively transitive if and only if there exists a partition $\{A_1, \ldots, A_m\}$ of $\mathcal{A}$ such that $(i, j) \in \mathcal{F}$ if and only if i and j belong to different sets in this partition.*

To see the induced structure, define $i \sim j$ if and only if $i = j$ or $(i, j) \notin \mathcal{F}$. Under symmetry and negative transitivity, $\sim$ is an equivalence relation, and its equivalence classes are the induced proximity clusters. Hence, the zone-coverage graph is complete multipartite: within-cluster pairs are inadmissible and all cross-cluster pairs are admissible.

We note that the partition structure characterized in Lemma 1 need not represent all zone-coverage constraints implied by a real geographical topology, which may induce intersecting clusters or vary across slots. In such environments, the partition representation and the resulting exact characterizations need not extend. Nevertheless, modeling zone-coverage as a negatively transitive binary relation $\mathcal{F}$ provides a tractable partition approximation.

We refer to the resulting partition $\{A_1, \ldots, A_m\}$ as the *(induced) clusters*. Whenever convenient, we refer to a generic cluster A_ℓ simply by its index ℓ. The cluster index of agent i is denoted by $[i]$; that is, $i \in A_{[i]}$. Throughout, $\mathcal{F}$ and its induced clusters are fixed.

Assumption 1 *The zone-coverage relation $\mathcal{F}$ is symmetric and negatively transitive, and $\{A_1, \ldots, A_m\}$ are the induced clusters.*

Whether the induced cluster structure admits a feasible allocation depends, in particular, on the capacity vector $q = (q_s)_{s \in \mathcal{S}}$. Feasibility immediately implies $q_s \leq m$ for each slot s. We refer to the subclass induced by Assumption 1 as *scheduling problems under cluster-approximated spatial distribution constraints*, or simply as *problems* hereafter.

3 Many-to-many Matching under Hard Quotas and Spatial Distribution Constraints

We begin with the special case of single-unit demands (Section 3.1), which highlights the main challenge and develops the core intuition for our analysis. We then extend the analysis to multi-unit demands and heterogeneous capacities (Section 3.2). While this general model builds on the same intuition, it introduces additional technical challenges, which we address in the Appendix.

3.1 Single-unit-demand Problem

We consider the simplest version of the problem. For spatial distribution constraints to bind, slots must have capacity at least two. Accordingly, we begin with the single-unit-demand case, in which each agent $i \in \mathcal{A}$ has demand $d_i = 1$ and each slot $s \in \mathcal{S}$ has capacity $q_s = 2$. The quota condition then implies $|\mathcal{A}| = 2|\mathcal{S}|$. We identify each singleton assignment with its unique slot.

3.1.1 Axioms.

We first introduce the standard notion of efficiency.

Definition 2 *A feasible allocation μ is **Pareto efficient** if there exists no feasible allocation μ' such that for each $i \in \mathcal{A}$, $\mu'(i)\ \pi_i^R\ \mu(i)$ and for at least one $j \in \mathcal{A}$, $\mu'(j)\ \pi_j^P\ \mu(j)$.*

We consider a fairness requirement that rules out justified envy. An allocation satisfies *no justified-envy* with respect to $\succ$ if a higher-priority agent who prefers a lower-priority agent's assignment could obtain that bundle only by harming an agent ranked above them. This property provides strict top-down protection of priority claims.

Definition 3 *An allocation μ satisfies **no justified-envy** under $\succ$ if for any $i \succ j$ with $\mu(j)\ \pi_i^R\ \mu(i)$, and any feasible μ' with $\mu'(i) = \mu(j)$, there exists $k \succ i$ such that $\mu(k)\ \pi_k^P\ \mu'(k)$.*

We next introduce a stability notion based on pairwise deviations. To this end, we extend the agents' priority ordering to pairs. Specifically, we represent any two–agent set $\{i, j\}$ by the ordered tuple $(i, j)_\succ$, where $i \succ j$. We then define an ordering $\succeq_c$ over agent pairs as

$$(i, j)_\succ \succeq_c (i', j')_\succ \text{ if and only if } i \succeq i' \text{ and } j \succeq j',$$

with strict part $(i,j)_\succ \succ_c (i',j')_\succ$ whenever at least one inequality is strict and $(i,j) \neq (i',j')$. We refer to this extension as *pair-priority ordering*.

Given a feasible allocation μ, a zone-covering pair (i,j) *blocks* μ if there exists $s \in \mathcal{S}$ such that $s\ \pi_i^P\ \mu(i)$ and $s\ \pi_j^P\ \mu(j)$, $(i,j) \in \mathcal{F}$, and $(i,j) \succ_c \mu_s$. A zone-covering pair (i,j) *weakly blocks* μ if there exists $s \in \mathcal{S}$ such that $s\ \pi_i^R\ \mu(i)$ and $s\ \pi_j^R\ \mu(j)$, $(i,j) \in \mathcal{F}$, and $(i,j) \succeq_c \mu_s$. The notion of stability follows from this definition of a two–agent deviation at a single slot.

Definition 4 *A feasible allocation μ is **pairwise** (resp. **strongly**) **stable** if no zone-covering pair (i,j) (resp. weakly) blocks μ.*

The definition is pair-based since the minimal feasible assignment to a slot is not a single agent but a zone-covering pair. A slot cannot evaluate an agent independently of her potential partner: even a high-priority agent can be assigned to a slot only together with another agent who satisfies the zone-coverage relation. Thus, the relevant local deviation is not that an individual agent joins a slot, but that an admissible pair of agents jointly replaces the current pair at that slot. Pairwise stability should therefore be read as the absence of locally admissible zone-covering-pair deviations. The notion is local and spatially constrained: it checks whether a single slot can be profitably recomposed by a zone-covering pair. A blocking deviation is not required to satisfy a separate global completion test. Adding such a requirement would restrict the set of blocking deviations and therefore yield a weaker stability notion; global completion is instead enforced by the mechanisms introduced below.

We finally define the incentive property used in the mechanism analysis.

Definition 5 *A mechanism φ is **strategy-proof** if there exist no agent $i \in \mathcal{A}$, preference profile $\pi = (\pi_i, \pi_{-i})$, and alternative report $\widehat{\pi}_i$ such that*

$$\mu_i^\varphi(\widehat{\pi}_i, \pi_{-i})\ \pi_i^P\ \mu_i^\varphi(\pi_i, \pi_{-i}).$$

3.1.2 Deferred Acceptance. We next analyze the well-known *deferred acceptance (DA)*. For each $A \subseteq \mathcal{A}$, define $F_A = \{(i,j)_\succ \in A \times A : (i,j) \in \mathcal{F}\}$.

Lemma 2 *For each $A \subseteq \mathcal{A}$, either $F_A = \emptyset$, or there exists $(i_A^*, j_A^*)_\succ \in F_A$ such that for each $(i,j)_\succ \in F_A$, $(i_A^*, j_A^*)_\succ \succeq_c (i,j)_\succ$. Moreover, in the latter, i_A^* is the $\succ$ − maximal agent in A.*

The pair-priority ordering induces a canonical choice from any agent set in $\mathcal{A}$. For each $A \subseteq \mathcal{A}$, if $F_A \neq \emptyset$, then by Lemma 2, (i_A^*, j_A^*) exists and i_A^* is the $\succ$-maximal agent in A. We define the *(priority-respecting) choice function* $C_s \colon 2^{\mathcal{A}} \to 2^{\mathcal{A}}$ as follows:

$$C_s(A) = \begin{cases} \{i_A^*, j_A^*\}, & \text{if } F_A \neq \emptyset, \\ \{i_A^*\}, & \text{if } F_A = \emptyset. \end{cases}$$

The transformation from an incomplete pair order to a choice function allows us to analyze how properties of $\mathcal{F}$ develop into desirable properties of C_s. We say, a choice function $C \colon 2^{\mathcal{A}} \to 2^{\mathcal{A}}$ is *substitutable* if for each $A \subseteq B \subseteq \mathcal{A}$ and $i \in A$, whenever $i \in C(B)$, then $i \in C(A)$.

Proposition 1 *For every symmetric and negatively transitive zone-coverage relation $\mathcal{F}$ and every priority ordering $\succ$, the priority-respecting choice function is single-valued and substitutable.*

The result has a matroidal interpretation. More generally, for each slot s with capacity q_s, let

$$\mathcal{I}_s = \{I \subseteq \mathcal{A} : |I| \leq q_s \text{ and } |I \cap A_\ell| \leq 1 \text{ for every } \ell = 1, \ldots, m\}$$

denote the family of locally admissible agent sets. Ignoring the overall capacity constraint, the restriction that at most one agent may be selected from each proximity cluster defines a partition matroid whose blocks are the clusters and whose block capacities are one. Imposing the additional bound $|I| \leq q_s$ gives its truncation at rank q_s. Equivalently, $(\mathcal{A}, \mathcal{I}_s)$ is the laminar matroid generated by the family $\{\mathcal{A}, A_1, \ldots, A_m\}$, with capacity q_s on $\mathcal{A}$ and capacity one on each cluster A_ℓ (Fife and Oxley, 2017).

In the quota-two environment considered here, the priority-respecting choice function defined above coincides with the fixed-priority greedy choice induced by this matroid: agents are considered in priority order and retained whenever their cluster is not yet represented, until the slot reaches

capacity or all applicants have been considered. Because the priority ordering is strict, this greedy choice is single-valued. Moreover, fixed-priority greedy choice under a matroid constraint is substitutable (Yokoi, 2019). We nevertheless provide a short, self-contained proof of Proposition 1 in the Appendix.

Single-valuedness makes each slot's tentative holding unique, while substitutability places the slot-level choices within the standard substitutable-choice domain used by DA.

Deferred Acceptance under priority ordering $\succ$ (DA).

Step 1 Each agent proposes to her most preferred slot. Each slot s considers the set $A_s \subseteq \mathcal{A}$ of proposers it has received and tentatively holds $C_s(A_s)$, rejecting all others.

Step k Each agent, rejected at Step $k-1$, proposes to her most preferred remaining slot. Each slot considers new proposals and stand-by agents from the previous steps, applies C_s to this combined set, and rejects others.

The process ends when no rejections occur. Each agent is assigned to the slot that holds it, and each slot holds exactly the set of agents chosen in the final step by C_s.

Substitutability of choice functions (Proposition 1) ensures that DA terminates in finitely many rounds. However, as the next example shows, the DA may not result in a feasible outcome even if there exists one.

Example 1 *Let $\mathcal{A} = \{i_1, i_2, i_3, i_4\}$ and $\mathcal{S} = \{s_1, s_2\}$. The set of zone-covering pairs is induced by the partition $\{A_1, A_2, A_3\} = (\{i_1, i_2\}, \{i_3\}, \{i_4\})$ such that $\mathcal{F} = \{(i, j) \in \mathcal{A} \times \mathcal{A} : [i] \neq [j]\}$. The priority ordering is given as $i_3 \succ i_4 \succ i_1 \succ i_2$. Let agents' preferences be $i_1 \colon s_2 \; \pi^P_{i_1} \; s_1$, $i_2 \colon s_2 \; \pi^P_{i_2} \; s_1$, $i_3 \colon s_1 \; \pi^P_{i_3} \; s_2$ and $i_4 \colon s_1 \; \pi^P_{i_4} \; s_2$. Observe that there are two feasible allocations: in the first, agents $\{i_1, i_3\}$ and $\{i_2, i_4\}$ are assigned to slots s_1 and s_2, respectively; in the second, agents $\{i_1, i_4\}$ and $\{i_2, i_3\}$ are assigned to slots s_1 and s_2, respectively. However, the outcome of the DA, namely the allocation μ such that $\mu_{s_1} = \{i_3, i_4\}$, and $\mu_{s_2} = \{i_1\}$, violates feasibility.*

Notice that each slot's holding is independent in its local matroid: $\{i_3, i_4\} \in \mathcal{I}_{s_1}$ and $\{i_1\} \in \mathcal{I}_{s_2}$. The failure is therefore not local. Agent i_2 is the only unmatched agent and belongs to the same cluster

as i_1, which is already represented at the only slot with residual capacity. Thus, for the singleton cluster set $L = \{1\}$, residual demand is one while residual effective capacity is zero.

This example motivates the following characterization for the DA's outcome being feasible.

Theorem 1 *Let $\mathcal{F}$ be symmetric and negatively transitive and let $\{A_1, \ldots, A_m\}$ be the induced proximity clusters. Suppose a feasible allocation exists. Then, for each $\succ$ and $(\pi_i)_{i \in \mathcal{A}}$, the DA outcome is feasible if and only if one of the following is true:*

1. *$m = 2$,*
2. *for each $\ell = 1, \ldots, m$, $|A_\ell| = 1$.*

Theorem 1 identifies two sources of DA's uniform feasibility guarantee. In the balanced case, $m = q = 2$, every full feasible slot contains one agent from each cluster, so completing a slot preserves the cluster balance required for completion. In the all-singleton case, the spatial constraint is non-binding because every pair of distinct agents is compatible. Apart from this all-compatible case, two clusters are necessary and sufficient for the uniform guarantee. When the constraint is non-vacuous and $m > 2$, locally admissible acceptances may alter residual cluster demands so that no feasible completion remains; DA does not test for this possibility. Appendix B extends the characterization to homogeneous quota q.

This motivates a mechanism with per-step global checks: the next subsection introduces a rule that accepts a tentative assignment only if the resulting partial allocation still admits a feasible completion.

3.1.3 Feasible Serial Dictatorship. We propose a version of the serial dictatorship mechanism that incorporates feasibility constraints. To this end, we introduce the notion of a partial allocation. A partial allocation specifies the assignments made up to a given step of the mechanism, subject to the requirement that all quota and zone-coverage conditions are respected.

Definition 6 *An allocation μ is **partial** if*

1. *For each $i \in \mathcal{A}$, $|\mu(i)| \leq 1$,*

2. *for each* $s \in \mathcal{S}$, $|\mu_s| \leq 2$,

3. $\mu_s = \{i, j\}$ *implies* $(i, j) \in \mathcal{F}$.

A partial allocation μ *admits a completion* if there exists a feasible allocation μ' such that for each $i \in \mathcal{A}$, $\mu(i) \subseteq \mu'(i)$. Each such allocation μ' is called a *completion*.

Let μ be an allocation. If $|\mu_s| = 1$, then we denote the unique agent in μ_s as $j_\mu(s)$. We denote the set of *vacant slots* by $S_0^\mu = \{s \in \mathcal{S} : |\mu_s| = 0\}$; for each $i \in \mathcal{A}$, the set of *partially filled slots* available for i by $S_1^\mu(i) = \{s \in \mathcal{S} : |\mu_s| = 1 \text{ and } (i, j_\mu(s)) \in \mathcal{F}\}$; the set of available slots for i by $S^\mu(i) = S_0^\mu \cup S_1^\mu(i)$; the set of available slots for agents in A by $S_A^\mu = \bigcup_{i \in A} S^\mu(i)$; and the set of agents in A_ℓ without an assigned slot under μ by $A_\ell(\mu) = \{i \in A_\ell : \mu(i) = \emptyset\}$.

A necessary condition for μ to admit a completion is $|A_\ell(\mu)| \leq |S_{A_\ell}^\mu|$ for every cluster ℓ (pigeonhole principle): otherwise, some two agents in A_ℓ must share a slot in any completion, violating the zone-coverage condition in Definition 1. The converse also holds:

Theorem 2 *A partial allocation μ admits a completion if and only if for each $\ell = 1, \ldots, m$, $|A_\ell(\mu)| \leq |S_{A_\ell}^\mu|$.*

We observe that under the empty allocation (no slots assigned), Theorem 2 reduces to $|A_\ell| \leq |\mathcal{S}|$ for each ℓ. Then, the quota condition ($|\mathcal{A}| = 2|\mathcal{S}|$) implies that each cluster contains at most half of all agents. We adopt this as a standing assumption for the remainder of the section.

For any partial allocation μ, we define $D_\ell(\mu) = |S_{A_\ell}^\mu| - |A_\ell(\mu)|$. By Theorem 2, there exists a feasible allocation if and only if for each ℓ, $D_\ell(\mu^0) \geq 0$. We define the set of *feasibly available* slots for i as follows:

$$S_{f-av}^\mu(i) = S_0^\mu \cup \left\{ s \in S_1^\mu(i) : \text{ for each } \ell \notin \{[i], [j_\mu(s)]\},\ D_\ell(\mu) > 0 \right\}.$$

We are now ready to introduce the serial dictatorship mechanism adjusted for feasibility, which refines the standard procedure to ensure that quota and zone-coverage constraints are respected.

Feasible Serial Dictatorship (FSD).

Step 0 Let μ^0 be the empty allocation.[3]

Step k Assign i_k to the π_{i_k} *maximal* element of $S_{f-av}^{\mu^{k-1}}(i_k)$. Denote the updated allocation by μ^k.

The mechanism FSD terminates at Step $k = |\mathcal{A}|$, and its outcome is the allocation $\mu^{|\mathcal{A}|}$. Our main result in this section shows that $\mu^{|\mathcal{A}|}$ is feasible, and satisfies all the axioms stated in Section 3.1.1.

Theorem 3 *For each $\succ$, FSD is strategy-proof. Moreover, for each $(\pi_i)_{i\in\mathcal{A}}$, the outcome of FSD satisfies feasibility, no justified-envy, pairwise stability and Pareto efficiency.*

3.2 Multi-unit-demand Problem with Heterogeneous Capacities

We extend the single-unit-demand model of Section 3.1 to a setting with heterogeneous multi-unit demands and heterogeneous slot capacities. The notion of partial allocation carries over: allocations must respect slot capacities q_s while satisfying the zone-coverage condition.

Construction of replica agents. The construction has two steps: we first expand each agent into unit-demand replicas and then order those replicas consistently with the original priority order. For each agent i, we create $|d_i|$ replicas. This construction is essential: once agents have multi-unit demands, the extension of the priority ordering $\succ$ directly affects the interpretation of the axioms. For each $i \in \mathcal{A}$, we construct *the set of replicas of* i by $\Gamma(i) = \{i^1, i^2, \ldots, i^{d_i}\}$, and for each $A \subseteq \mathcal{A}$, the set of all replicas of agents in A is denoted by $A^\Gamma = \bigcup_{i\in A} \Gamma(i)$. We endow $\mathcal{A}^\Gamma$ with the replica order $\succ^\Gamma$, defined by:

$$i^k \succ^\Gamma j^\ell \text{ if and only if } k < \ell \text{ or } \big(k = \ell \text{ and } i \succ j\big).$$

In the replica order $\succ^\Gamma$, (i) each agent's first replica precedes each agent's second replica, each second replica precedes each third replica, and so forth, and (ii) within each layer of replicas with the same index k, that is, the kth replicas of all agents, the ordering is identical to the priority ordering $\succ$. For example, consider three agents i, j, k, each demanding two units, with priority ordering $i \succ j \succ k$. Under our construction, the induced replica order $\succ^\Gamma$ is given by the following. $i^1 \succ^\Gamma j^1 \succ^\Gamma k^1 \succ^\Gamma i^2 \succ^\Gamma j^2 \succ^\Gamma k^2$.[4]

[3] Note that, as mentioned earlier, we impose that for each ℓ, $D_\ell(\mu^0) \geq 0$ such that the quota condition is satisfied.

[4] Note that an alternative replica ordering, say $\succ^\Delta$, could be that $i^1 \succ^\Delta i^2 \succ^\Delta j^1 \succ^\Delta j^2 \succ^\Delta k^1 \succ^\Delta k^2$. Under such an

We choose the replica order $\succ^\Gamma$ to satisfy two minimal consistency requirements: (i) respect for $\succ$—a lower-priority agent cannot overtake a higher-priority one merely because they demand fewer units; and (ii) symmetry across units—no agent's first unit is disadvantaged relative to another's later unit. Ordering replicas lexicographically by (replica index, then $\succ$) satisfies both: all agents compete on equal footing for their first unit before any second unit is considered, preventing high-demand agents from crowding out low-demand agents early in the allocation.

We extend the zone-coverage relation to replicas by defining $\mathcal{F}^\Gamma \subseteq \mathcal{A}^\Gamma \times \mathcal{A}^\Gamma$ such that:

$$(i^k, j^\ell) \in \mathcal{F}^\Gamma \text{ if and only if } i \neq j \text{ and } (i, j) \in \mathcal{F}.$$

Observe that if $\mathcal{F}$ is negatively transitive, so is $\mathcal{F}^\Gamma$; the induced replica clusters are $A_1^\Gamma, \ldots, A_m^\Gamma$, where A_ℓ^Γ contains all replicas of agents in A_ℓ.

Residual demand. Let μ be a partial allocation. The set of replicas of agents in A_l without an assigned slot under μ is denoted by $A_\ell^\Gamma(\mu) = \{i^k \in A_\ell^\Gamma : [i] = \ell \text{ and } \mu(i^k) = \emptyset\}$. Then, for each $L \subseteq \{1, \ldots, m\}$, the total residual demand is $\sum_{\ell \in L} |A_\ell^\Gamma(\mu)|$.

Residual supply. The availability of a slot to an agent in cluster ℓ is constrained by both its remaining capacity and the clusters of the agents assigned to the slot. Let $q_s^\mu = q_s - |\mu(s)|$ be the residual capacity of s at μ. For each s, let $L_s^\mu \subseteq \{1, \ldots, m\}$ be the subset of clusters such that $\ell \in L_s^\mu$ if and only if there exists no i^k such that $i^k \in \mu_s$ and $[i] = \ell$. Thus, the *effective capacity* (or *residual supply*) of s for the agents in clusters in L is $\min(|L_s^\mu \cap L|, q_s^\mu)$.

For illustration, suppose that a slot s has one unit of residual capacity and already contains a replica from cluster 1. Then $q_s^\mu = 1$ and cluster 1 is excluded from L_s^μ. If $L = \{1, 2\}$ and cluster 2 remains admissible at s, the slot contributes $\min(|L_s^\mu \cap L|, q_s^\mu) = 1$ unit of effective capacity for the replicas in clusters 1 and 2. Thus, residual capacity alone does not determine availability; the clusters already represented at the slot also matter.

order, an agent with larger demand may exhaust available capacity before lower-priority agents receive any assignment, even when a feasible allocation exists that assigns at least one unit to each agent.

Theorem 4 *A partial allocation μ admits a completion if and only if for each $L \subseteq \{1, \ldots, m\}$,*

$$\sum_{\ell \in L} |A_\ell^\Gamma(\mu)| \leq \sum_{s \in \mathcal{S}} \min(|L_s^\mu \cap L|, q_s^\mu).$$

The proof of Theorem 4 rests on a graph-theoretic reduction and two classical results: the Ford–Fulkerson max-flow/min-cut theorem and the integrality theorem for network flows. In Appendix A, we construct an auxiliary directed graph whose nodes and arc capacities encode the supply, demand, zone-coverage constraints, and admissible completions; feasible completions of the original problem correspond bijectively to integral flows on this graph. Applying the Ford–Fulkerson theorem yields necessary and sufficient conditions for the existence of a feasible (possibly fractional) flow, and the integrality theorems guarantee that, under the capacity structure we impose, there exists a feasible flow that is integral. These two steps together produce the characterization stated in the theorem.

The inequalities in Theorem 4 need not be verified by enumerating all subsets of clusters. As established by the maximum-flow construction in its proof, a partial allocation admits a completion if and only if the corresponding residual network has a flow whose value equals total residual demand. Thus, completion feasibility can be tested by a single maximum-flow computation. The same completion condition also has a matroidal interpretation as a matroid-union problem over slot-specific residual matroids; we discuss this connection in Appendix C.

Theorem 4 requires that for any subset of clusters, the number of unassigned replicas does not exceed the aggregate effective capacity of the slots.[5] Consequently, a feasible allocation exists only if the empty allocation μ^0 satisfies:

$$\sum_{\ell \in L} |A_\ell^\Gamma(\mu^0)| \leq \sum_{s \in \mathcal{S}} \min(|L_s^{\mu^0} \cap L|, q_s^{\mu^0}).$$

Note that at μ^0, $A_\ell^\Gamma(\mu^0)$ is the set of all replicas in cluster ℓ, and $L_s^{\mu^0}$ is the set of all clusters. Moreover, for each s, $q_s^{\mu^0} = q_s$. These imply the following corollary of Theorem 4.

[5] Observe that when for each s, $q_s = 2$ and $|L| = 1$, this inequality collapses to the one in Theorem 2.

Corollary 1 *A feasible allocation exists if and only if for each $L \subseteq \{1, \ldots, m\}$, we have*

$$\sum_{\ell \in L} \sum_{i \in A_\ell} d_i \leq \sum_{s \in \mathcal{S}} \min(|L|, q_s).$$

The broader implication is that, under hard demand and capacity constraints, feasibility hinges critically on the specification of the zone-coverage relation—and thus on the structure of the induced clusters. In our pharmacy-scheduling application in Türkiye, these hard constraints bind because duty shifts are a major revenue source and the institution prioritizes each pharmacy's centrally set demand. Corollary 1 therefore implies that ensuring an allocation satisfying both demand and capacity constraints requires careful design of the zone-coverage relation. In practice, however, the central authority fixes relatively dispersed subsets of pharmacies to meet hard demand and capacity constraints and reuses these subsets across shifts. The data, however, indicate that this heuristic occasionally fails the zone-coverage requirement (Section 4). Absent the formal treatment suggested by Corollary 1, many allocations that would be feasible under the same zone-coverage relation are effectively excluded, mirroring manual subset selection.

We maintain the condition in Corollary 1 as a standing assumption for the remainder of the analysis. Fixing a zone-coverage relation that (by assumption) guarantees feasibility, we assign agents to slots dynamically via local decisions, while preserving global feasibility at every step and ensuring efficiency, fairness, and stability.[6] To this end, we first define the set of available slots for i at μ:

$$S^\mu(i) = \{s \in \mathcal{S} : q_s^\mu > 0 \text{ and } [i] \notin \{[k] : k \in \mu(s)\}\}.$$

To prevent a structural deadlock, we define the *feasibly available* slots for i at μ as follows:

$$S^\mu_{f-av}(i) = \{s \in S^\mu(i) : \forall\, L \subseteq \{1, \ldots, m\},\ \sum_{\ell \in L} |A^\Gamma_\ell(\mu \cup \{(i,s)\})| \leq \sum_{s' \in \mathcal{S}} \min(|L^{\mu \cup \{(i,s)\}}_{s'} \cap L|,\ q^{\mu \cup \{(i,s)\}}_{s'})\}.$$

[6] In our running application, this approach replaces the central authority's heuristic in Türkiye with a principled, feasibility-guaranteeing method: rather than fixing specific subsets of pharmacies, we fix a zone-coverage relation and treat feasibility formally by selecting pharmacies for shifts systematically and dynamically.

By Theorem 4, a slot s belongs to $S^{\mu}_{f-av}(i)$ if and only if the tentative partial allocation $\nu = \mu \cup \{(i, s)\}$ admits a feasible completion. The displayed inequalities provide an analytical characterization of this condition. In the implementation of $GFSD$, however, membership in $S^{\mu}_{f-av}(i)$ is evaluated by the maximum-flow test described after Theorem 4, rather than by enumerating all subsets L. Equivalently, a locally available slot remains a candidate precisely the tentative assignment of a replica to that slot leaves the residual problem completable.

That is, assigning i to a feasibly available slot ensures that for each subset of clusters L, the remaining demand does not exceed the remaining effective capacity. The mechanism processes the replicas sequentially according to the order $\succ^{\Gamma}$. We are now ready to define the mechanism we propose.

Generalized Feasible Serial Dictatorship ($GFSD$).

Step 0 Let μ^0 be the empty allocation.

Step k Consider the highest-priority replica under $\succ^{\Gamma}$ that has not yet been processed, and assign it to its most preferred slot in its feasibly available set. Denote the updated allocation by μ^k.

The mechanism $GFSD$ terminates at Step $k = |\mathcal{A}^{\Gamma}|$, and its outcome is the allocation $\mu^{|\mathcal{A}^{\Gamma}|}$. For each $i \in \mathcal{A}$, $\mu(i) = \mu(i^1) \cup \ldots \cup \mu(i^{d_i})$.

Proposition 2 *On the cluster-approximated domain, $GFSD$ can be implemented in time polynomial in the size of the original instance.*

With multi-unit demand, π_i orders individual slots. We extend it to bundles by listing each bundle's slots from best to worst under π_i and comparing bundles at the first rank where they differ, hence the *lexicographic extension*: For any allocation μ and each $i \in \mathcal{A}$, index the elements of $\mu(i)$ as $\mu_i^{(1)}, \ldots, \mu_i^{(d_i)}$ such that $\mu_i^{(1)} \ \pi_i \ \mu_i^{(2)} \ \pi_i \ \cdots \ \pi_i \ \mu_i^{(d_i)}$. Agent i *lexicographically prefers* $\mu(i)$ to $\mu'(i)$, denoted $\mu(i) \ \pi_i^{\text{lex}} \ \mu'(i)$, if there exists $t \in \{1, \ldots, d_i\}$ such that:

$$\mu_i^{(t)} \ \pi_i \ \mu_i'^{(t)} \text{ and for each } r < t, \ \mu_i^{(r)} = \mu_i'^{(r)}.$$

The lexicographic extension is parsimonious in an informational sense. The primitive preference data consist only of each agent's ordinal ranking over individual slots. Alternative bundle models may

introduce additional structure: additive representations require cardinal values or weights, responsiveness imposes restrictions on how slot rankings extend to bundles, and complementarity-based models specify interactions across slots. By contrast, the lexicographic extension orders each bundle from best to worst according to π_i and compares bundles at the first rank where they differ. It therefore introduces no cardinal utilities, weights, an additive-separability assumption, or parameters governing substitution or complementarity across slots. Thus, it is not the weakest possible extension in a normative sense, but a transparent complete ranking convention based only on the primitive slot ranking.

Since every replica has unit demand, the no justified-envy requirement in Definition 3 applies directly to the replica market. Specifically, we replace $\mathcal{A}$ and $\succ$ in that definition with $\mathcal{A}^\Gamma$ and $\succ^\Gamma$, respectively; each replica i^r inherits the slot preference π_i of its agent; and feasible alternatives are evaluated with respect to replica feasibility.[7]

Because demands are hard, the existence of such a feasible allocation μ' implies $d_i = d_j$, so the relevant bundle comparison is between bundles of equal cardinality. We now define the standard efficiency criterion with respect to the lexicographic extension.

Definition 7 *An allocation μ is **Pareto efficient** if there exists no feasible allocation μ' such that for each $i \in \mathcal{A}$, $\mu'(i)\ \pi_i^{\text{lex}}\ \mu(i)$ or $\mu'(i) = \mu(i)$ and for at least one $j \in \mathcal{A}$ $\mu'(j)\ \pi_j^{\text{lex}}\ \mu(j)$.*

Throughout the multi-unit analysis, Pareto efficiency and incentive comparisons are evaluated under the lexicographic extension. No justified-envy and pairwise stability are formulated separately at the replica level under the inherited slot preferences and the replica priority order $\succ^\Gamma$.

Remark 1 *The lexicographic extension is a complete binary relation and it is weaker than the standard responsive extension. Therefore, Pareto efficiency is stronger under lexicographic extension than the extension of responsive preferences.*[8]

[7] No justified-envy can alternatively be formulated by comparing agents' complete bundles under the lexicographic extension. $GFSD$ does not satisfy this bundle-level property in general; Appendix D provides the formal definition and a counterexample.

[8] Under the responsive extension, agent i prefers $\mu(i)$ to $\mu'(i)$ if the sets $\mu(i)$ and $\mu'(i)$ differ at only one slot, $s \in \mu(i)$ and $s' \in \mu'(i)$, respectively, such that $s\ \pi_i^P\ s'$.

We also define pairwise stability analogously as before, replacing $\mathcal{F}$ with $\mathcal{F}^\Gamma$ and using the same pair-priority rule on replicas. Formally, under quota two, a zone-covering replica pair (i^r, j^ℓ) blocks an allocation μ at slot s if $s \; \pi_i^P \; \mu(i^r)$, $s \; \pi_j^P \; \mu(j^\ell)$, $(i^r, j^\ell) \in \mathcal{F}^\Gamma$, and $(i^r, j^\ell)_{\succ^\Gamma} \succ_c \mu_s$. An allocation is pairwise stable at the replica level if no such pair exists. We are now ready to state our main result.

Theorem 5 *For each $\succ^\Gamma$ and each $(\pi_i)_{i\in\mathcal{A}}$, the outcome of $GFSD$ satisfies feasibility, no justified-envy, pairwise stability, and Pareto efficiency.*

In contrast to the single-unit FSD, $GFSD$ is not strategy-proof in general. The following example shows that the replica construction can create incentives for an agent to misreport.

Example 2 *Let $\mathcal{A} = \{i_1, i_2\}$ and $\mathcal{S} = \{s_1, s_2, s_3\}$. Slot capacities are $q_{s_1} = q_{s_2} = q_{s_3} = 1$. Agent demands are $d_{i_1} = 2$ and $d_{i_2} = 1$. Since each slot has capacity one, the zone-coverage constraint is irrelevant for this example. Let the priority ordering be $i_1 \succ i_2$. Then the set of replicas is $A^\Gamma = \{i_1^1, i_2^1, i_1^2\}$, and the replica order induced by $\succ^\Gamma$ is $i_1^1 \succ^\Gamma i_2^1 \succ^\Gamma i_1^2$.*

Let agent i_1's true preference be $s_1 \; \pi_{i_1}^P \; s_2 \; \pi_{i_1}^P \; s_3$, and let agent i_2's preference be $s_2 \; \pi_{i_2}^P \; s_3 \; \pi_{i_2}^P \; s_1$. Each replica inherits the preference of its agent. Under truthful reporting, i_1^1 chooses s_1, then i_2^1 chooses s_2, and finally i_1^2 receives s_3. Thus, agent i_1 obtains the bundle $\{s_1, s_3\}$.

Now suppose that agent i_1 instead reports $s_2 \; \widehat{\pi}_{i_1}^P \; s_1 \; \widehat{\pi}_{i_1}^P \; s_3$. Under this report, i_1^1 chooses s_2, then i_2^1 chooses s_3, and finally i_1^2 receives s_1. Thus, agent i_1 obtains the bundle $\{s_1, s_2\}$.

Under agent i_1's true preference and the lexicographic extension, $\{s_1, s_2\}$ is strictly preferred to $\{s_1, s_3\}$. Hence agent i_1 has a profitable misreport, and $GFSD$ is not strategy-proof in general.

The example isolates the source of the incentive failure. Since all slots have capacity one, the zone-coverage constraint plays no role. Manipulability arises instead from the interaction between multi-unit demand and sequential allocation: an early replica of an agent can change the opportunity set faced by a later replica of the same agent. Thus, truth-telling should not be expected from $GFSD$ in the unrestricted multi-unit domain. We therefore interpret $GFSD$ as a completion-preserving allocation rule under reported or estimated preferences, rather than as a dominant-strategy implementation.

4 Pharmacy Duty-shift Scheduling: An Application

We evaluate the duty-shift scheduling problem in Section 1 using a data-driven simulation, comparing institutional practice with our mechanism on welfare and assignment structure. We use administrative records from Ankara, Türkiye (2015–2025): 3,905 shifts and 213 distinct pharmacies in the raw panel. After restricting attention to pharmacies with complete coordinate, priority, and cluster-recovery information, the simulation sample contains 198 pharmacies, with each record listing the shift, assigned pharmacies, and geolocations. All counterfactual mechanisms use the 198-pharmacy sample drawn from the Keçiören district of Ankara. Figure 3 depicts the 188 pharmacies with positive 2024 duty demand, whereas Figure 5 depicts the full 198-pharmacy cluster-recovery sample. Accordingly, the reported tables and figures should not be interpreted as province-wide representations of Ankara. Shifts carry revenue coefficients (relative profitability reported by the Istanbul Chamber of Pharmacists, ICP), and capacities vary across the calendar, generating heterogeneous workloads across pharmacies.

Institutional practice does not formalize zone-coverage or priority; districts rely on informal partner sets and rotations that only partially reflect coverage. Coverage is implemented through shift-specific adjustments rather than a stable zone-coverage relation.[9] While co-assignments are typically dispersed, Table 1 documents a number of ultra-proximate pairings in 2019–2025: 16 distinct pairs are assigned jointly at distances below 500 meters, appearing a total of 33 times; 75% of these occur only once, and the minimum observed distance is 51 meters.[10] At the same time, one pair appears 10 times, consistent with the authority repeatedly reusing a small set of ‘‘known-feasible’’ exceptions to meet hard demand and capacity constraints. This motivates reconstructing zone-coverage from observed co-assignments and proximity before evaluating counterfactual mechanisms.

For counterfactual evaluation, we reconstruct primitives from data: infer zone-coverage by combining observed co-assignments with geographic proximity; generate preferences via a utility model using official duty-shift coefficients plus idiosyncratic noise; match multi-unit demand and shift ca-

[9] Across all co-assigned pharmacy pairs (pair-days), the median distance is 3.44 km and the 90th percentile is 5.54 km (2015–2025).

[10] These failures of the zone-coverage requirement underscore the importance of Corollary 1 and its explicit incorporation into the allocation mechanism.

Table 1: Ultra-proximate co-assignments under the Status Quo schedule

Threshold	Unique pairs	Pair-days	Share once (%)	Max count	Min dist. (m)
< 500m	16	33	75.0	10	51

Notes: A pair-day is a shift-day in which two pharmacies are co-assigned. ''Share once'' is the fraction of unique pairs that appear exactly once in the panel. ''Max count'' is the largest number of appearances of any single pair below 500m. ''Min dist.'' is the minimum observed distance among co-assignments below 500m.

pacities to their empirical values; and construct priorities from cumulative duty-shift coefficients (2015–2023). These steps yield a fully specified simulation environment that preserves the observed scheduling structure while enabling systematic counter-factuals. Reconstruction details are given in Appendix F.

4.1 Simulation Results

We evaluate four mechanisms in the reconstructed environment:

- the *Status Quo (SQ)*, the manually constructed schedule by Ankara Chamber of Pharmacists,
- $GFSD$, our proposed mechanism that allows any cross cluster partnership
- *Fixed-Pair* $GFSD$, the $GFSD$ under the SQ's admissible partner graph, and
- *Myopic Serial Dictatorship (MSD)*, a benchmark that ignores feasibility constraints.

The performance of these mechanisms hinges—beyond normative axioms—on the induced assignment structure. Restrictive admissibility narrows interaction patterns and limits slack propagation. We refer to this dimension as *structural flexibility*—the breadth of realized partnerships versus reliance on a small core—measured by total partners and core partners. These metrics proxy cross-cluster slack propagation, required to satisfy Theorem 4. We compare these mechanisms on feasibility, welfare, and structural flexibility. The robustness grid combines eight cluster-recovery or geographic specifications with five preference calibrations, $\rho \in \{0.0, 0.3, 0.5, 0.7, 0.9\}$, yielding 40 scenarios. The eight specifications consist of five repeated co-duty thresholds, $t \in \{2, 3, 5, 7, 10\}$, and three alternative geographic criteria evaluated relative to the baseline recovery.

- *Feasibility.* *SQ* achieves full assignment via discretionary adjustments. By contrast, hard-coding its partner admissibility into Fixed-Pair $GFSD$ assigns, on average, only 86.9% of replicas, indicating that such rigidity is incompatible with systematic priority-based allocation. Across 40 scenarios, MSD produces 1.78 unassigned replicas on average (violating the condition in Theorem 4), whereas $GFSD$ clears the market without failures.Thus, the main feasibility conclusion is robust: $GFSD$ is the only mechanism that uniformly clears the market across the full scenario grid.

  The gap between MSD and $GFSD$ varies across preference environments. MSD failures remain small on average, yet they persist in some scenarios because myopic assignments need not preserve feasible completion. The gap is the largest under more heterogeneous preference environments, thus under low correlation, preferences, where MSD is most prone to deadlocks, while $GFSD$ clears the market for each preference profile. Further evidence is reported in Appendix E.

Table 2: Average Mechanism Performance Across Robustness Scenarios

Mechanism	**Yield** (%)	**Unassigned**	**Mean Rank**	**Median Rank**	**Top-3** (%)	**Partners** (≥ 1)	**Core** (≥ 2)
Status Quo	100.0	0.00	181.4	180.0	1.2	13.7	3.9
Fixed-Pair GFSD	86.9	206.70	100.9	61.9	9.1	11.0	6.6
MSD	99.9	1.78	**29.0**	**26.1**	**15.7**	28.0	**2.1**
$GFSD$	**100.0**	**0.00**	29.3	**26.1**	15.7	**28.1**	2.1

Notes: Entries report averages over the full robustness grid, consisting of alternative cluster reconstructions, alternative distance specifications, and alternative preference calibrations (40 scenarios in total). Welfare metrics for all mechanisms, including the status quo, are computed under the same simulated preference profiles; they should therefore be interpreted as model-implied counterfactual welfare measures. Observed welfare under the status quo is unavailable because preferences were not elicited. For mechanisms with unassigned replicas, rank-based welfare statistics are computed conditional on assigned replicas; unassigned counts are reported separately. Lower rank indicates better preference satisfaction. Total partners count all unique co-assignees (≥ 1), while core partners reflect persistent connections (≥ 2).

- *Welfare.* Pharmacy preferences are not elicited, so realized welfare under the status quo is not observed. The *SQ* allocation is observed, however, and we evaluate it under the same simulated preference profiles used for the counterfactual mechanisms. The reported rank-based statistics are therefore model-implied counterfactual welfare measures, not observed welfare outcomes.

  Under this common preference scale, *SQ* performs poorly: its average rank is 181.4, its median rank is 180.0, and only 1.2% of assignments fall within the top three preferred duty shifts. Fixed-Pair

$GFSD$ also yields a poor performance (mean rank 100.9, median 61.9). The large gap between mean and median evidences a right-skewed distribution. Rigid admissibility forces a nontrivial fraction of pharmacies into extremely undesirable duty shifts. Both MSD and $GFSD$ generate superior outcomes, with nearly 16% of assignments falling in the top three preferred duty shifts. On average, their welfare levels are nearly identical on the assigned margin. MSD generates, on average, 1.78 unassigned replicas due to violations of the feasibility condition in Theorem 4. These failures arise in instances where demand concentrates on a small set of high-score duty shifts with limited capacity: early greedy assignments exhaust local slack and preclude feasible continuation. By contrast, $GFSD$ explicitly guards against such feasibility-destroying moves through its look ahead feasibility check and therefore clears the market without failures.

- *Structural flexibility.* The SQ exhibits moderate partner diversity (13.7) but relies on a rigid backbone of persistent partnerships (3.9 core partners). Fixed-Pair $GFSD$ performs worse: the admissibility restriction forces repeated use of a small feasible backbone (6.6 core partners). In contrast, $GFSD$ integrates the market: total partner diversity rises to 28.1 while core partnerships fall to 2.1. This pattern—high diversity with low persistence—is the structural signature of market integration.

  The averages in Table 2 mask variation across reconstructed environments, which we summarize in Appendix E.1. There, we report ranges and standard deviations across the robustness grid, including alternative repeated co-duty thresholds, distance specifications, and preference environments. The qualitative ranking of mechanisms remains unchanged throughout: $GFSD$ clears the market in every tested scenario, whereas MSD leaves positive numbers of replicas unassigned in some scenarios and Fixed-Pair $GFSD$ remains structurally constrained.

5 Further Applications

We present several applications in health services that fall squarely within our theoretical framework.

- *Home Health Care Workforce Scheduling.*

  Home-health agencies assign nurses or carers to daily shifts that involve visiting patients at fixed addresses across metropolitan or rural regions. Feasibility depends not only on workload and travel

time but on geographic-dispersion requirements: each shift must cover a subset of patient locations that collectively ensure service across predefined zones (e.g., north/south districts, urban/rural sectors) to meet response-time targets, equity goals, or contractual obligations. Caregivers express preferences over shifts, neighborhoods, patient types, and times—some preferring compact routes, others specific districts—so the assignment must jointly form geographically feasible bundles of visits and match them to workers in a way that respects both coverage constraints and ordinal preferences; this structure corresponds to shifts as admissible zone-covering subsets and sites (or workers) with ordinal preferences over assignments. (Mascolo, Martinez, and Espinouse, 2021)

- *Community and Public Health Outreach Services.*

  Public health departments operate fixed clinics and outreach points across socio-economically distinct neighborhoods, and shift planning must ensure each service period includes sufficiently dispersed sites to guarantee equitable access—especially for vulnerable populations. Thus admissible shifts are governed by a geographic coverage structure: a feasible shift must represent multiple designated zones so no area is systematically underserved. Clinics and service teams also have preferences over days, times, travel distance, and community specialization, so scheduling becomes a matching problem with structured feasibility: only zone-covering combinations of sites form shifts, and among those feasible combinations, bilateral preferences determine the final allocation. (Jin, 2023)

- *Emergency Medical Services (EMS) and Ambulance Deployment.*

  Emergency medical systems operate from a network of fixed stations across a city or region, and each shift must position ambulances and paramedic teams so coverage is geographically balanced and response times meet mandated limits. This creates explicit or implicit zone-coverage constraints: a shift is feasible only if the selected stations collectively cover all critical districts, so clustering units in one area can violate coverage elsewhere even when overall staffing is adequate. Paramedics express preferences over station assignments and shift types (day/night, central/peripheral), and stations may prefer particular team compositions or continuity; the allocation must therefore reconcile

dispersion-based feasibility with bilateral preferences, making this a natural instance of a geographically constrained matching problem. (Li, Zhao, and Wyatt, 2011)

- *Multi-Facility Care Home or Hospital Staff Scheduling.*

  In large urban systems with multiple long-term care homes or hospitals across city zones, staff scheduling must ensure each shift staffs a geographically dispersed subset of facilities—especially when floating or rotating personnel are used to stabilize service—so feasible assignments are governed by coverage rules that require minimum representation across zones to meet equity and regulatory standards. Nurses and other healthcare workers rank preferred shifts and locations (commuting, team familiarity, workload), and facilities express preferences (experienced staff, continuity of care), making scheduling a preference-sensitive matching problem in which shifts are zone-covering bundles and both sides hold preferences over feasible assignments. (Anderson, Bodur, Rathwell, and Sarhangian, 2021)

6 Concluding Remarks

This paper studies scheduling problems in which cluster-approximated spatial distribution requirements and hard capacity constraints fundamentally shape the set of feasible allocations. In such environments, feasibility is not a local property that can be enforced slot by slot, but a global condition that depends on how assignments interact across clusters of agents and slots. As shown both theoretically and through application, this feature causes standard matching mechanisms—designed around local acceptance or greedy priority rules—to fail even when feasible allocations exist. To address this problem, we propose a priority-based mechanism ($GFSD$) that embeds a dynamic global feasibility check into the assignment process. The mechanism guarantees full assignment whenever feasibility is attainable while preserving priority respect, no justified-envy, Pareto efficiency and pairwise stability. Importantly, the construction accommodates heterogeneous demands and capacities without sacrificing these normative properties.

Our empirical analysis of pharmacy duty-shift scheduling demonstrates the practical relevance of these results. Using administrative data to reconstruct the scheduling environment, we show that rigid

admissibility rules and myopic priority mechanisms can generate assignment failures or poor model-implied rank outcomes under the simulated preference profiles. By enforcing generalized feasibility conditions, the proposed mechanism preserves full assignment in every reported scenario and improves model-implied rank outcomes without altering priorities or the maintained preference profiles. These gains arise from structural flexibility, highlighting the importance of mechanisms that enable coordinated allocation across clusters. More broadly, our results establish a general design principle for constrained matching and scheduling problems in which feasible completion admits a tractable characterization.

Appendix A Proofs

Proof of Lemma 1. (Only if) Suppose $\mathcal{F}$ is symmetric and negatively transitive. Define a binary relation $\sim$ on $\mathcal{A}$ such that $i \sim j$ if and only if $i = j$ or $(i, j) \notin \mathcal{F}$. We show that $\sim$ is an equivalence relation. By definition, for each $i \in \mathcal{A}$, $i \sim i$ holds. Hence, $\sim$ is reflexive. Moreover, since $\mathcal{F}$ is symmetric, $(j, i) \in \mathcal{F}$ implies $(i, j) \in \mathcal{F}$. By definition of $\sim$, if $i \sim j$ and $i \neq j$, then $(i, j) \notin \mathcal{F}$. Thus, if $i \sim j$, then $j \sim i$. Hence, $\sim$ is symmetric. Lastly, if $i \sim j$ and $j \sim k$, and $i \neq j$, $j \neq k$, then $(i, j) \notin \mathcal{F}$ and $(j, k) \notin \mathcal{F}$. By negative transitivity of $\mathcal{F}$, $(i, k) \notin \mathcal{F}$. Thus, $i \sim k$, implying transitivity of $\sim$. Thus, $\sim$ is an equivalence relation. Let $\{A_1, \ldots, A_m\}$ be its equivalence classes, that form a partition of $\mathcal{A}$, and $[i]$ denote the $\sim$ *class* of i. We claim that $(i, j) \in \mathcal{F}$ if and only if $[i] \neq [j]$. If $[i] = [j]$ and $i \neq j$, then $i \sim j$ implies $(i, j) \notin \mathcal{F}$; if $i = j$, the same conclusion follows from irreflexivity. Conversely, if $[i] \neq [j]$, then $i \neq j$ and $i \nsim j$, which implies $(i, j) \in \mathcal{F}$. Hence $\mathcal{F}$ is exactly the set of cross-class pairs.

(If) Suppose there exists a partition $\{A_1, \ldots, A_m\}$ of $\mathcal{A}$ such that $(i, j) \in \mathcal{F}$ if and only if $[i] \neq [j]$. Symmetry of $\mathcal{F}$ follows from the definition. For negative transitivity, let $(i, j) \notin \mathcal{F}$ and $(j, k) \notin \mathcal{F}$. Then $[i] = [j]$ and $[j] = [k]$, hence $[i] = [k]$, which implies $(i, k) \notin \mathcal{F}$. ■

Proof of Lemma 2. Fix $A \subseteq \mathcal{A}$ and suppose $F_A \neq \emptyset$. Let i^*_A be the $\succ$–maximal agent in the set $\{\, i \in A : \text{there exists } j \in A \text{ such that } (i, j)_\succ \in F_A \,\}$ and let j^*_A be the $\succ$–maximal agent such that $(i^*_A, j^*_A)_\succ \in F_A$. Now take any $(i, j)_\succ \in F_A$. By the choice of i^*_A, we have $i^*_A \succeq i \succeq j$. If $(i^*_A, j)_\succ \in F_A$, by choice of j^*_A, $j^*_A \succeq j$. Thus, for each $(i, j)_\succ \in F_A$ we have $i^*_A \succeq i$ and $j^*_A \succeq j$, which is equivalent to $(i^*_A, j^*_A)_\succ \succeq_c (i, j)_\succ$. ■

Proof of Proposition 1. Let $\mathcal{F}$ be negatively transitive and $\{A_1, \ldots, A_m\}$ be the induced clusters. If $F_A = \emptyset$, i^*_A is unique. If $F_A \neq \emptyset$, Lemma 2 implies F_A has a $\succeq_c$ − maximal pair $(i^*_A, j^*_A)_\succ$, hence $C_s(A)$ is uniquely determined. For substitutability, let $A \subseteq B$ and $i \in A$. Suppose $i \in C_s(B)$. We show $i \in C_s(A)$. If $i = i^*_B$, then it implies $i = i^*_A$. Thus, $i \in C_s(A)$. If $F_B \neq \emptyset$ and $i = j^*_B$, then there are two cases. First, suppose $i^*_B \in A$. In this case, i remains the highest–priority zone-covering partner to i^*_B in A. Thus, $i \in C_s(A)$. Second, suppose $i^*_B \notin A$. Since, by Lemma 2, $i = j^*_B$ is the $\succ$–maximal agent

of $B \setminus \{i_B^*\}$ whose cluster differs from $[i_B^*]$, any $k \in B$ with $k \succ i$ must satisfy $[k] = [i_B^*]$; otherwise k would be a strictly higher priority zone-covering partner of i_B^*, contradicting $i = j_B^*$. In particular, for each $k \in A$ with $k \succ i$ we have $[k] = [i_B^*]$. Thus, $[i_A^*] = [i_B^*] \neq [i]$, and, $(i_A^*, i) \in \mathcal{F}$, and among $A \setminus \{i_A^*\}$ the agent i is the $\succ$–maximal agent whose cluster different from $[i_A^*]$. Hence, $i = j_A^*$ and $i \in C_s(A)$. ■

Proof of Theorem 1. Since every slot has quota two and must contain agents from distinct clusters, the existence of a feasible allocation implies $m \geq 2$. Suppose $m = 2$. Let A_1, A_2 be the clusters. Since a feasible allocation exists and each filled slot must be cross-class by Lemma 1, $|A_1| = |A_2| = |\mathcal{S}|$. Suppose DA allocation is infeasible. Note that the priority-respecting choice function chooses two agents from a set only if there exists a zone-covering pair. Thus, infeasibility arises only if some slots are under-filled. Let i be the first agent in the priority ordering who remains unmatched. Agent i eventually proposes to every slot. At the end of DA, each slot is therefore either full or holds a single agent in $A_{[i]}$; otherwise, a slot with residual capacity and an incumbent from the other cluster would retain i as a compatible second agent. Hence, every slot contains an agent from $A_{[i]}$. Since each slot can accommodate at most one agent from a given cluster, this implies that all $|\mathcal{S}|$ agents in $A_{[i]}$ are matched. Because $|A_{[i]}| = |\mathcal{S}|$, no agent in $A_{[i]}$ can remain unmatched, contradicting that i is unmatched.

Suppose next that every cluster is a singleton. Then every pair of distinct agents is zone-covering, so the spatial constraint never binds and the priority-respecting choice function reduces to the standard quota-two priority choice rule. If an agent were unmatched at the end of DA, she would have been rejected by every slot, so every slot would hold two agents. The slots would then hold all $2|S| = |A|$ agents, contradicting the existence of an additional unmatched agent.

Suppose $m > 2$ and that at least one cluster is non-singleton. Let M be a largest cluster and choose distinct agents $i_1, i_2 \in M$. Since a feasible allocation exists, $|M| \leq |S|$. Moreover, every cluster other than M has size at most $|S| - 1$: if another cluster also had size $|S|$, those two clusters would contain all $2|S|$ agents, contradicting $m > 2$. After removing i_1 and i_2, there remain $2(|S| - 1)$ agents, with no cluster represented more than $|S| - 1$ times. These agents can therefore be partitioned into $|S| - 1$

zone-covering pairs. Indeed, a multiset of $2r$ cluster labels can be partitioned into r pairs of distinct labels whenever no label occurs more than r times; the result follows by repeatedly pairing agents from two currently largest clusters. Let these pairs be $B_1, \ldots, B_{|S|-1}$, and let the slots be $s_0, s_1, \ldots, s_{|S|-1}$. For each $k \geq 1$, both agents in B_k rank s_k first. Agents i_1 and i_2 rank s_0 first, and i_2 ranks the remaining slots after s_0 in any strict order. Complete all other preferences arbitrarily, and choose a common priority ordering in which i_2 has the lowest priority and $i_1 \succ i_2$. In the first round, each s_k, $k \geq 1$, holds the pair B_k, while s_0 holds i_1 and rejects i_2 because i_1 and i_2 belong to the same cluster. Agent i_2 is then rejected by every remaining slot, each of which already holds a zone-covering pair of higher-priority agents. DA terminates with i_2 unmatched and one vacant position at s_0. Thus, although a feasible allocation exists, DA is not guaranteed to produce one. ■

Proof of Theorem 2. (If) Let $\mathcal{M}$ be the set of all partial allocations that do not admit a completion such that for each $\ell = 1, \ldots, m$ and $\mu \in \mathcal{M}$, $|A_\ell(\mu)| \leq |S^\mu(A_\ell)|$. Define $\mu \in \mathcal{M}$ such that:

$$\mu \in \arg\max_{\mu' \in \mathcal{M}} \sum_{i \in \mathcal{A}} |\mu'(i)| \tag{1}$$

Let μ^f be a feasible allocation. Then, the assumption $|\mathcal{A}| = 2|\mathcal{S}|$, and the quota condition $|\mu^f_s| = 2$ for each s imply for each $i \in \mathcal{A}$, $\mu^f(i) \neq \emptyset$. Since μ is a partial allocation, $\mu_s = \{i, j\}$ implies $(i, j) \in \mathcal{F}$. Thus, there are $s \in \mathcal{S}$ and $i \in \mathcal{A}$ such that $|\mu_s| = 1$ and $\mu(i) = \emptyset$. That is, $i \in A_{[i]}(\mu)$. Since $|A_{[i]}(\mu)| \leq |S^\mu(A_{[i]})|$, there exists at least one $s \in \mathcal{S}$ such that $s \in S^\mu(A_{[i]})$.

Let μ' be an allocation such that for each $s' \neq s$, $\mu_{s'} = \mu'_{s'}$ and $\mu_{s'} = \mu_s \cup \{i\}$. Note that $|A_{[i]}(\mu')| \leq |S^{\mu'}(A_{[i]})|$ still holds and there is no change for other clusters. Thus, μ' is a partial allocation, and $\sum_{i \in \mathcal{A}} |\mu'(i)| > \sum_{i \in \mathcal{A}} |\mu(i)|$, which contradicts (1).

(Only if) Let μ be a partial allocation that admits a completion and μ' be a feasible allocation such that for each $i \in \mathcal{A}$, $\mu(i) \subseteq \mu'(i)$. Let ℓ be an index such that $|A_\ell(\mu)| > |S^\mu(A_\ell)|$. That is, for at least one $i \in A_\ell(\mu)$, there is no slot s such that $j_\mu(s)$ is a zone-covering partner for i. This contradicts the existence of feasible μ'. ■

Proof of Theorem 3. Recall that $i_1 \succ i_2 \succ \ldots \succ i_{|\mathcal{A}|}$, and for each ℓ, $D_\ell(\mu^0) \geq 0$ (an assumption

such that a feasible allocation exists–see the footnote in the description of the FSD mechanism in Section 3.1.3). Let μ be the outcome of FSD.

Feasibility. By Theorem 2, it is enough to show that after each step k at FSD, for each ℓ, $D_\ell(\mu^k) \geq 0$. We argue this by induction. At $k = 0$, the inequality follows directly from the characterizing condition for the existence of a feasible allocation restated above, establishing the initial step of the induction.

Inductive step. For each ℓ, let $D_\ell(\mu^{k-1}) \geq 0$. Thus, μ^{k-1} admits a completion. Hence, i_k is assigned to some slot in at least one completion, implying $S^{\mu}_{f-av}(i_k) \neq \emptyset$. We assign i_k to her most preferred slot s in $S^{\mu}_{f-av}(i_k)$. We consider two cases of s being vacant or partially filled.

Suppose $s \in S_0^{\mu^{k-1}}$. Then, for cluster $A_{[i_k]}$, $|A_{[i_k]}(\mu)| = |A_{[i_k]}(\mu - 1)| - 1$ and $|S^{\mu^k}_{f-av}(i_k)| = |S^{\mu^{k-1}}_{f-av}(i_k)| - 1$. That is, there is one agent less without an assigned slot and slot s is no longer available to agents in $A_{[i_k]}$. Thus, $D_{[i_k]}(\mu^k) \geq 0$. For cluster $\ell \neq [i_k]$, the assignment of i_k does not affect the set of agents in A_ℓ without an assigned slot, and while slot s moves from vacant to partially filled, the set $S^{\mu}_{A_\ell}$ of available slots remains unchanged; hence $D_\ell(\mu^k) = D_\ell(\mu^{k-1}) \geq 0$.

Suppose $s \in S_1^{\mu^{k-1}}(i_k)$. Then, $(i_k, j_{\mu^{k-1}}(s)) \in \mathcal{F}$ and $|\mu^k_s| = 2$. For cluster $[i_k]$, $|A_{[i_k]}(\mu)| = |A_{[i_k]}(\mu^{k-1})| - 1$ and $|S^{\mu^k}_{f-av}(i_k)| = |S^{\mu^{k-1}}_{f-av}(i_k)| - 1$. Thus, $D_{[i_k]}(\mu^k) = D_{[i_k]}(\mu^{k-1}) \geq 0$. For cluster $[j_{\mu^{k-1}}(s)]$, there is no change in both the set of agents without an assigned slot and the set of available slots for all agents in $A_{[j_{\mu^{k-1}}(s)]}$. For clusters $\ell \neq \{[i_k], [j_{\mu^{k-1}}(s)]\}$, there is no change in the set of agents without an assigned slot, but the set of available slots for all agents in A_ℓ is reduced by one. Thus, $D_\ell(\mu^k) = D_\ell(\mu^{k-1}) - 1$. Since $s \in S_1^{\mu^{k-1}}(i_k)$, the set of feasibly available slots for i_k requires that for each $\ell \neq \{[i_k], [j_{\mu^{k-1}}(s)]\}$, $D_\ell(\mu^{k-1}) > 0$. Thus, $D_\ell(\mu^k) \geq 0$.

No justified-envy. Let μ be the outcome of FSD and μ' be a feasible allocation. Suppose μ does not satisfy no justified-envy. There exists $k' > k$ such that $s = \mu(i_{k'})\ \pi^R_{i_k}\ \mu(i_k)$. Also, $\mu'(i_k) = s$ and for each $k'' < k$, $\mu'(i_{k''})\ \pi^R_{i_{k''}}\ \mu(i_{k''})$. Thus, $s \notin S^{\mu^{k-1}}_{f-av}(i_k)$ otherwise i_k would be assigned to s at Step k. Consider the partial allocation $\tilde{\mu}$ such that for each $k'' \leq k$, $\tilde{\mu}(k'') = \mu'(k'')$ and for each $k' > k$, $\tilde{\mu}(k') = \emptyset$. Since μ' is feasible, $\tilde{\mu}$ admits a completion. Thus, for at least one $k'' < k$, $\mu(k'') \neq \mu'(k'')$. Otherwise, $s \in S^{\mu^{k-1}}_{f-av}(i_k)$.

Let $l = \min_{l'}\{l' : l' < k \text{ and } \mu'(i_{l'}) \neq \mu(i_{l'})\}$. Since μ does not satisfy no justified-envy, $\mu'(i_l)\ \pi^P_{i_l}$

$\mu(i_l)$. Thus, $\mu'(i_l) \notin S_{f-av}^{\mu^{l-1}}(i_l)$ otherwise i_l would be assigned to $\mu'(i_l)$ at Step l. It implies either for at least one $l' < l$, $\mu(l') \neq \mu'(l')$ or $l = 1$. Since l is defined to be the highest priority agent with $\mu'(i_l) \neq \mu(i_l)$, we have $l = 1$. This yields $\mu'(i_1)\ \pi_{i_1}^P\ \mu(i_1)$, contradicting that $\mu(i_1)$ is the $\pi_{i_1} - maximal$ element of $\mathcal{S}$.

Pairwise stability. Suppose, toward a contradiction, that $(i, j)_\succ$ blocks the FSD outcome μ at slot s, where $i \succ j$. Because both blocking agents strictly prefer s to their assignments, neither belongs to μ_s. Pair-priority dominance therefore implies that i has higher priority than every agent in μ_s. Since FSD processes agents in priority order and never revokes an assignment, slot s was empty when i was processed. Hence $s \in S_0^{\mu^{k-1}}$. Since i strictly prefers s to $\mu(i)$, this contradicts the choice made by FSD.

Pareto efficiency. Suppose μ is not Pareto efficient. Then, there is a feasible μ' such that for each $i \in \mathcal{A}$, $\mu'(i)\ \pi_i^R\ \mu(i)$, and for at least one $j \in \mathcal{A}$, $\mu'(j)\ \pi_j^P\ \mu(j)$. Let k^* be the smallest index in the priority ordering such that $\mu(i^{k^*}) \neq \mu'(i^{k^*})$. Thus, for each $k < k^*$, $\mu(i^k) = \mu'(i^k)$. Since agents have strict preference ordering over slots, $\mu(i^{k^*})\ \pi_{i^{k^*}}^R\ \mu'(i^{k^*})$ implies either $\mu(i^{k^*})\ \pi_{i^{k^*}}^P\ \mu'(i^{k^*})$ or $\mu(i^{k^*}) = \mu'(i^{k^*})$. Since $\mu(i^{k^*}) \neq \mu'(i^{k^*})$, $\mu'(i^{k^*})\ \pi_{i^{k^*}}^P\ \mu(i^{k^*})$.

Since μ' is feasible and coincides with μ^{k^*-1} on all higher-priority agents, $\mu^{k^*-1} \cup \{(i^{k^*}, \mu'(i^{k^*}))\}$ admits a feasible completion. Thus, $\mu'(i^{k^*}) \in S_{f-av}^{\mu^{k^*-1}}(i^{k^*})$. But, FSD assigns i^{k^*} at Step k^* to a $\pi_{i^{k^*}}$-maximal element of $S_{f-av}^{\mu^{k^*-1}}(i^{k^*})$. Since $\mu'(i^{k^*})\ \pi_{i^{k^*}}\ \mu(i^{k^*})$, it leads a contradiction.

Strategy-proofness. Fix an agent i_k and fix the reports of all agents other than i_k. Let π_{i_k} be the true preference of i_k, and let $\widehat{\pi}_{i_k}$ be an arbitrary alternative report. Let μ^t denote the partial allocation after Step t when i_k reports truthfully, and let $\widehat{\mu}^t$ denote the partial allocation after Step t when i_k reports $\widehat{\pi}_{i_k}$, holding all other reports fixed.

We first show that the two runs coincide up to Step $k - 1$. At Step 0, both runs start from the empty allocation, so $\mu^0 = \widehat{\mu}^0$. Suppose that $\mu^{t-1} = \widehat{\mu}^{t-1}$ for some $t < k$. The agent processed at Step t is $i_t \neq i_k$, and this agent has the same reported preference in both runs. Moreover, the set of feasibly available slots $S_{f-av}^{\mu^{t-1}}(i_t)$ is determined only by the current partial allocation and by the primitive feasibility data, namely the cluster structure, capacities, and remaining unassigned agents.

It does not depend on the reported preference of i_k. Hence

$$S_{f-av}^{\mu^{t-1}}(i_t) = S_{f-av}^{\widehat{\mu}^{t-1}}(i_t).$$

Therefore, i_t chooses the same slot in both runs, and $\mu^t = \widehat{\mu}^t$. By induction, $\mu^{k-1} = \widehat{\mu}^{k-1}$.

It follows that when i_k is processed, the set of feasibly available slots is the same under truthful reporting and under the alternative report. Let this common set be

$$B_{i_k} = S_{f-av}^{\mu^{k-1}}(i_k) = S_{f-av}^{\widehat{\mu}^{k-1}}(i_k).$$

Under truthful reporting, FSD assigns i_k to the π_{i_k}-maximal element of B_{i_k}. Under the alternative report $\widehat{\pi}_{i_k}$, FSD assigns i_k to the $\widehat{\pi}_{i_k}$-maximal element of the same set B_{i_k}. Since the truthful assignment is the best element of B_{i_k} according to i_k's true preference, the assignment obtained under any alternative report is weakly worse according to π_{i_k}.

Finally, because this is a single-unit-demand environment, the assignment of i_k is fixed at Step k and cannot be changed by later steps of the mechanism. Thus, there exists no alternative report $\widehat{\pi}_{i_k}$ such that i_k obtains a strictly better assignment than under truthful reporting. By Definition 5, FSD is strategy-proof. ■

Proof of Theorem 4. (Only if) Let μ be a partial allocation admitting a completion μ'. Fix $L \subseteq \{1, \ldots, m\}$ and s. At most q_s^μ replicas can be assigned to s due to capacity. Also, the zone-coverage allows at most one replica from each cluster in $L \cap L_s^\mu$. Thus, s can accommodate at most $\min(|L_s^\mu \cap L|, q_s^\mu)$ replicas from $\bigcup_{\ell \in L} A_\ell^\Gamma(\mu)$ at μ'. Summing over all slots yields:

$$\sum_{\ell \in L} |A_\ell^\Gamma(\mu)| \leq \sum_{s \in \mathcal{S}} \min(|L_s^\mu \cap L|, q_s^\mu).$$

(If) For each $L \subseteq \{1, \ldots, m\}$, suppose $\sum_{\ell \in L} |A_\ell^\Gamma(\mu)| \leq \sum_{s \in \mathcal{S}} \min\{|L_s^\mu \cap L|, q_s^\mu\}$. To demonstrate that μ has a completion, we employ tools from graph theory. We construct a directed graph $G = (V, E)$ where V is the set of vertices and E is the set of edges. V consists of a source σ, a sink τ, for each

$\ell \in \{1, \ldots, m\}$, a cluster vertex v_ℓ, and for each $s \in \mathcal{S}$, a slot vertex v^s:

$$V = \{\sigma, \tau\} \cup \{v_\ell : \ell \in \{1, \ldots, m\}\} \cup \{v^s : s \in \mathcal{S}\},$$

and E consists of an edge from σ to each v_ℓ, an edge from v_ℓ to v^s if and only if cluster ℓ is admissible at s, and an edge from each v^s to τ:

$$E = \{(\sigma, v_\ell) : \ell \in \{1, \ldots, m\}\} \cup \{(v_\ell, vs) : \ell \in L_s^\mu \text{ and } s \in \mathcal{S}\} \cup \{(v^s, \tau) : s \in \mathcal{S}\}.$$

Let $k : A \to \mathbb{R}_+$ be the capacity function which associates each $e = (v, v') \in E$ of G a nonnegative real number $k(a)$, called the capacity of edge e. The capacities of edges of G are defined as $k(\sigma, v_\ell) = |A_\ell^\Gamma(\mu)|$, $k(v_\ell, v^s) = 1$, and $k(v^s, \tau) = q_s^\mu$. The network constructed in this way is depicted in Figure 1. A σ–τ *flow* is a function $f : E \to \mathbb{R}_+$ satisfying for each $e \in E$, $0 \leq f(e) \leq k(e)$, and flow

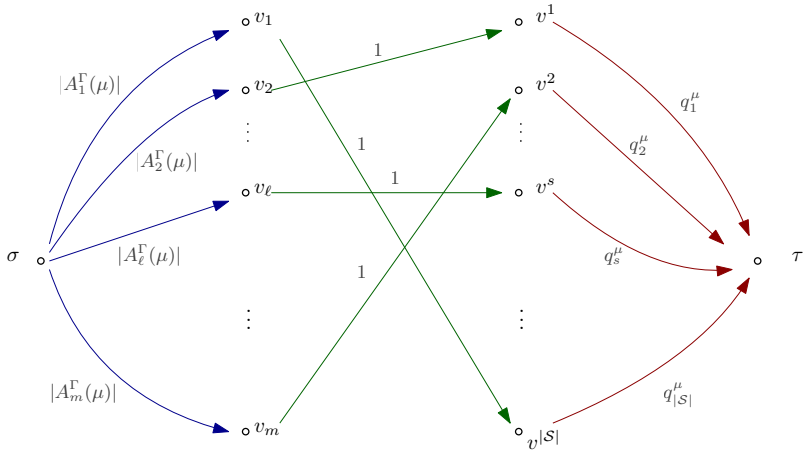


Figure 1

conservation such that for each $v \in V \setminus \{\sigma, \tau\}$: $\sum_{(v',v)\in E} f(v', v) = \sum_{(v,v'')\in E} f(v, v'')$. The *value* of a flow is the total flow sent out of σ. A partition of $V = U \cup \bar{U}$ is a $(\sigma, \tau) - cut$ if $\sigma \in U$ and $\tau \in \bar{U}$. Given a $(\sigma, \tau) - cut$ $(U, \bar{U})$, let $E' \subseteq E$ define the set of edges such that for each $e = (v, v') \in E'$, $v \in U$ and $v' \in \bar{U}$. We define the capacity of the cut as $k(U, \bar{U}) = \sum_{e\in E'} k(e)$. For a cut $(U, \bar{U})$, let

$S(U) = \{s : v^s \in U\}$, and $L = \{\ell : v_\ell \in U\}$. Note that the capacity of the cut $(U, \bar{U})$ is equal to

$$k(U, \bar{U}) = \sum_{\ell \notin L} |A_\ell^\Gamma(\mu)| + \sum_{s \in \mathcal{S} \setminus S(U)} |L_s^\mu \cap L| + \sum_{s \in S(U)} q_s^\mu.$$

To prove our result, it suffices to demonstrate the existence of a flow of value $T = \sum_{\ell=1}^m |A_\ell^\Gamma(\mu)|$ in the graph we constructed. This is achieved using the well-known Ford-Fulkerson Theorem, which characterizes the value of the maximum flow.

Theorem 6 *(Ford-Fulkerson Theorem) Let $G = (V, A)$ be a directed graph. The maximum value of the σ–τ flow is equal to the minimum capacity among all $(\sigma, \tau) - cut$.*

For each slot s, if $v^s \in U$ then the cut contributes q_s^μ, while if $v^s \in \bar{U}$ it contributes $|L_s^\mu \cap L|$; in both cases the contribution is at least $\min\{|L_s^\mu \cap L|,\ q_s^\mu\}$. That is,

$$\sum_{s \in \mathcal{S}} \min\{|L_s^\mu \cap L|, q_s^\mu\} \leq \sum_{s \in \mathcal{S} \setminus S(U)} |L_s^\mu \cap L| + \sum_{s \in S(U)} q_s^\mu. \tag{2}$$

Recall that we suppose for each L, $\sum_{\ell \in L} |A_\ell^\Gamma(\mu)| \leq \sum_{s \in \mathcal{S}} \min\{|L_s^\mu \cap L|, q_s^\mu\}$. By combining this and (2) and adding $\sum_{\ell \notin L} |A_\ell^\Gamma(\mu)|$ to both sides, we obtain

$$\underbrace{\sum_{\ell \in L} |A_\ell^\Gamma(\mu)| + \sum_{\ell \notin L} |A_\ell^\Gamma(\mu)|}_{T} \leq \underbrace{\sum_{s \in \mathcal{S} \setminus S(U)} |L_s^\mu \cap L| + \sum_{s \in S(U)} q_s^\mu + \sum_{\ell \notin L} |A_\ell^\Gamma(\mu)|}_{k(U, \bar{U})}$$

We now identify the minimum cut in Figure 2. Let $T = \sum_{\ell=1}^m |A_\ell^\Gamma(\mu)|$ denote the total residual demand for the set of all clusters. Thus, $k(U, \bar{U}) \geq T$. Since $(U, \bar{U})$ is arbitrary, the minimum capacity among all $(\sigma, \tau) - cut$ is greater than or equal to T. Consider the cut $(\{\sigma\}, V \setminus \{\sigma\})$. The only edges leaving σ are (σ, u_ℓ) such that:

$$k(\{\sigma\}, V \setminus \{\sigma\}) = \sum_{\ell \in \{1, \ldots, m\}} k(\sigma, u_\ell) = \sum_{\ell \in \{1, \ldots, m\}} |A_\ell^\Gamma(\mu)| = T.$$

Thus, the minimum cut's capacity is at most T. By the two bounds, the minimum cut capacity is

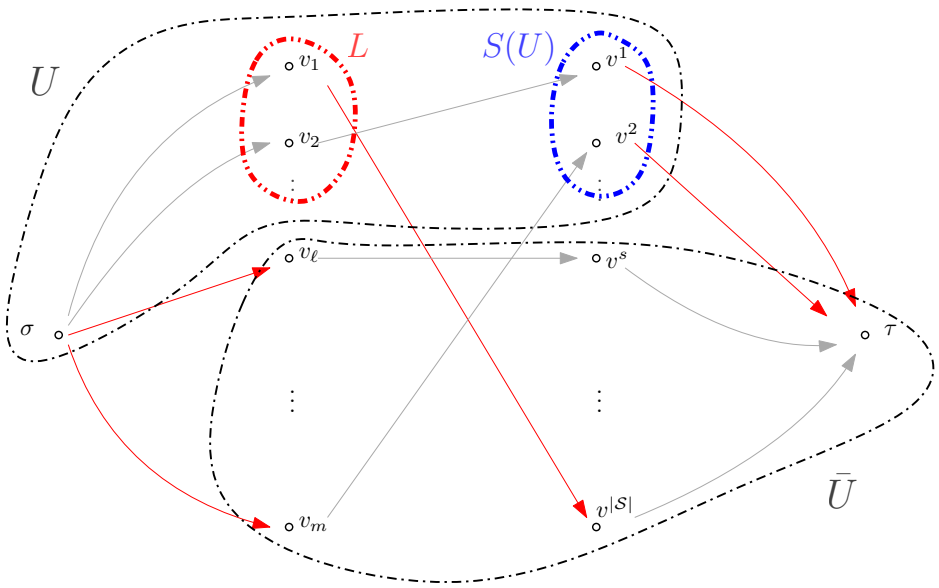


Figure 2

precisely T. By Theorem 6, the maximum value of σ—τ flow is therefore T. To establish an integer flow, we invoke the following classical theorem from combinatorics (Schrijver, 2003).

Theorem 7 *(Integer Theorem) Let $G = (V, A)$ be a directed graph. If for each $a \in A$, $k(a)$ is integer, then there exists a maximum σ–τ flow that is integral on every edge.*

By Theorem 7, there is an integer σ-τ flow of value T. We interpret each unit of flow along the path $\sigma \to v_\ell \to v^s \to \tau$ as assigning one replica of cluster ℓ to s. The capacity constraints on edges (σ, v_ℓ) ensure that exactly $|A_\ell^\Gamma(\mu)|$ replicas of cluster ℓ are assigned in total, while the capacities on edges (v^s, τ) ensure that no slot s receives more than its residual capacity. Finally, the unit capacities on edges (v_ℓ, v^s) guarantee that no slot receives more than one replica from the same cluster. Thus, the flow induces a completion of μ. ■

Proof of Proposition 2. Let $R = |\mathcal{A}^\Gamma| = \sum_{i \in \mathcal{A}} d_i$ denote the total number of replicas. Since a feasible allocation exists, for each $i \in \mathcal{A}$, $d_i \le |\mathcal{S}|$. Hence, $R = \sum_{i \in \mathcal{A}} d_i \le |\mathcal{A}||\mathcal{S}|$, thus the expanded replica representation has size polynomial in the original instance. Feasibility also implies $q_s \le m \le |\mathcal{A}|$ for every $s \in \mathcal{S}$, so this conclusion continues to hold when demands and capacities are encoded in binary. The mechanism processes each replica exactly once and therefore terminates after R assignment steps. At each step, it tests at most $|\mathcal{S}|$ candidate slots.

By the residual-network construction in the proof of Theorem 4, each candidate-slot test can be carried out by a single maximum-flow computation. The corresponding network contains one source, one node for each of the m clusters, one node for each slot, and one sink. It therefore has $m + |\mathcal{S}| + 2$ vertices. It has m source-to-cluster arcs, at most $m|\mathcal{S}|$ cluster-to-slot arcs, and $|\mathcal{S}|$ slot-to-sink arcs. Hence, it has at most $m + m|\mathcal{S}| + |\mathcal{S}|$ arcs.

Thus, $GFSD$ performs at most $R|\mathcal{S}|$ maximum-flow computations. Residual cluster demands, residual slot capacities, and cluster-presence indicators can be maintained throughout the mechanism, so each residual network can be constructed in $O(m|\mathcal{S}| + m + |\mathcal{S}|)$ time. Let $T_{\mathrm{MF}}(V, E)$ denote the running time of a strongly polynomial maximum-flow algorithm on a network with V vertices and E arcs. The running time of $GFSD$ is therefore

$$O(R|\mathcal{S}|\,[m|\mathcal{S}| + m + |\mathcal{S}| + T_{\mathrm{MF}}(m + |\mathcal{S}| + 2,\ m|\mathcal{S}| + m + |\mathcal{S}|)])\,.$$

Since $R \le |\mathcal{A}||\mathcal{S}|$ and $m \le |\mathcal{A}|$, this bound is polynomial in the size of the original instance. Hence, $GFSD$ is polynomial-time implementable on the cluster-approximated domain. ■

Proof of Theorem 5. Fix $\succ^{\Gamma}$ and recall that $i^1 \succ^{\Gamma} i^2 \succ^{\Gamma} \ldots \succ^{\Gamma} i^{|\mathcal{A}^{\Gamma}|}$. For each $k = 0, \ldots, |\mathcal{A}^{\Gamma}|$, let μ^k denote the Step k output of $GFSD$, with the empty allocation μ^0.

Feasibility. μ^0 satisfies the condition in Theorem 4, hence admits a completion. We prove by induction that, for each $k = 1, \ldots, |\mathcal{A}^{\Gamma}|$, μ^k admits a completion.

Inductive step. Assume μ^{k-1} satisfies the condition in Theorem 4, and hence admits a completion $\tilde{\mu}$. Let $\tilde{s}$ denote the slot to which replica i^k is assigned under $\tilde{\mu}$. Since $\tilde{\mu}$ is feasible and is a completion of μ^{k-1}, assigning i^k to $\tilde{s}$ and retaining the assignments of μ^{k-1}, respect both slot capacities and zone-coverage. Construct ν by assigning i^k to $\tilde{s}$ and retaining the remaining assignments of μ^{k-1} without reassignment. For cluster $[i^k]$, $|A_{[i^k]}(\nu)| = |A_{[i^k]}(\mu^{k-1})| - 1$. Moreover, $|\nu_{\tilde{s}}| = |\mu^{k-1}_{\tilde{s}}| + 1$. Thus, $q^{\nu}_{\tilde{s}} = q^{\mu^{k-1}}_{\tilde{s}} - 1$ and since $[i^k] \notin L^{\nu}_{\tilde{s}}$, for each $L \subseteq \{1, \ldots, m\}$, $|L^{\nu}_{\tilde{s}} \cap L| = |L^{\mu^{k-1}}_{\tilde{s}} \cap L| - 1$. Thus, $D_{[i_k]}(\nu) = D_{[i_k]}(\mu^{k-1}) \ge 0$ and the inequality of Theorem 4 continues to hold at ν. It follows that assigning i^k to $\tilde{s}$ preserves the completion criterion, and hence $\tilde{s} \in S^{\mu^{k-1}}_{f-av}(i^k)$. Thus, the feasibly

available slots for i^k is non-empty.

At Step k, the mechanism assigns i^k to a slot $s \in S^{\mu^{k-1}}_{f-av}(i^k)$ that is maximal under π_i and defines $\mu^k = \mu^{k-1} \cup \{(i^k, s)\}$. For cluster $[i^k]$, $|A_{[i^k]}(\mu^k)| = |A_{[i^k]}(\mu^{k-1})| - 1$. Moreover, $|\mu^k_s| = |\mu^{k-1}_s| + 1$. Thus, $q^{\mu^k}_s = q^{\mu^{k-1}}_s - 1$ and since $[i^k] \notin L^{\mu^k}_s$, for each $L \subseteq \{1, \ldots, m\}$, $|L^{\mu^k}_s \cap L| = |L^{\mu^{k-1}}_s \cap L| - 1$. Thus, $D_{[i_k]}(\mu^k) = D_{[i_k]}(\mu^{k-1}) \geq 0$ and the inequality of Theorem 4 continues to hold at μ^k. Hence μ^k admits a completion.

By induction, $\mu^{|\mathcal{A}^\Gamma|}$ admits a completion. Since the mechanism assigns exactly one slot to each replica and never revokes assignments, $\mu^{|\mathcal{A}^\Gamma|}$ is itself a feasible allocation.

No justified envy. Suppose toward a contradiction that the replica allocation μ produced by $GFSD$ violates no justified-envy. Then, there exist replicas $(i^r, j^\ell)_{\succ^\Gamma}$ with $\mu(j^\ell)\ \pi^P_i\ \mu(i^r)$ and a feasible replica allocation $\hat{\mu}$ with $\hat{\mu}(i^r) = \mu(j^\ell)$ such that, for each $h^q \succ^\Gamma i^r$, $\hat{\mu}(h^q)\ \pi^P_h\ \mu(h^q)$ or $\hat{\mu}(h^q) = \mu(h^q)$

Let k^t be the $\succ^\Gamma$-maximal replica in

$$\{i^r\} \cup \left\{h^q \in \mathcal{A}^\Gamma : h^q \succ^\Gamma i^r \text{ and } \hat{\mu}(h^q) \neq \mu(h^q)\right\}.$$

By the choice of k^t, each replica with priority higher than k^t is assigned identically under μ and $\hat{\mu}$. Moreover, $\hat{\mu}(k^t)\ \pi^P_k\ \mu(k^t)$. Indeed, if $k^t = i^r$, this follows from $\hat{\mu}(i^r) = \mu(j^\ell)\ \pi^P_i\ \mu(i^r)$. If $k^t \succ^\Gamma i^r$, it follows from the fact that $\hat{\mu}(k^t) \neq \mu(k^t)$ and that no replica with priority higher than i^r is made worse off under $\hat{\mu}$.

Consider the step of $GFSD$ at which k^t is processed. Since all higher-priority replicas are assigned identically under μ and $\hat{\mu}$, the assignments made before this step coincide with the corresponding assignments under $\hat{\mu}$. Because $\hat{\mu}$ is feasible, assigning k^t to $\hat{\mu}(k^t)$ while retaining all earlier assignments admits $\hat{\mu}$ as a feasible completion. Hence, $\hat{\mu}(k^t)$ belongs to the feasibly available set of k^t at the step at which k^t is processed.

Pairwise stability. Suppose that $(i^n, j^l)_{\succ^\Gamma}$ blocks $\mu^{|\mathcal{A}^\Gamma|}$. Thus, $(i, j) \in \mathcal{F}$ and there is s such that $s\ \pi^P_i\ \mu^{|\mathcal{A}^\Gamma|}(i^n)$, $s\ \pi^P_j\ \mu^{|\mathcal{A}^\Gamma|}(j^l)$, and $(i^n, j^l)_{\succ^\Gamma} \succ_c \mu_s$. Since $(i^n, j^l)_{\succ^\Gamma} \succ_c \mu_s$, the agent with the highest priority among $\mu_s \cup \{i^n, j^l\}$ is i^n. Because $GFSD$ processes replicas in the order $\succ^\Gamma$ and never revokes an assignment, none of the lower-priority replicas in μ_s has been assigned to s when i^n is pro-

cessed. By feasibility of the deviation, assigning i^n alone to s while fixing all higher-priority replicas preserves both local feasibility and the inequality of Theorem 4. Thus, for each $k < n$, $s \in S^{\mu^k}_{f-av}(i^n)$. Since $s\ \pi_i^P\ \mu^{|\mathcal{A}^\Gamma|}(i^n)$, $GFSD$ would have assigned i^n to s. Hence no blocking pair exists.

Pareto efficiency. Suppose the μ of $GFSD$ outcome μ is not Pareto efficient. Then, there is a feasible allocation μ' such that (i) for each $i \in \mathcal{A}$, either $\mu'(i) = \mu(i)$ or $\mu'(i)\ \pi_i^{\text{lex}}\ \mu(i)$, and (ii) there exists $j \in \mathcal{A}$ such that $\mu'(j)\ \pi_j^{\text{lex}}\ \mu(j)$. Let i^* be the agent with highest priority such that $\mu'(i^*) \neq \mu(i^*)$. Thus, for each $k \succ i^*$ we have $\mu'(k) = \mu(k)$. Since $\mu'(i^*)\ \pi_{i^*}^{\text{lex}}\ \mu(i^*)$, there exists a smallest index $t \in \{1, \ldots, d_{i^*}\}$ such that $\mu_{i^*}^{(t)'}\ \pi_{i^*}\ \mu_{i^*}^{(t)}$ and for each $r < t$, $\mu_{i^*}^{(r)'} = \mu_{i^*}^{(r)}$.

Consider the step at which the replica i^{*t} is processed. μ^{k-1} coincides with μ' on all assignments involving replicas with higher priority than i^{*t} in $\succ^\Gamma$. This holds for all replicas of agents $k \succ i^*$ by the choice of i^*, and it holds for the first $t-1$ replicas of i^* since for each $r < t$ $\mu_{i^*}^{(r)'} = \mu_{i^*}^{(r)}$. Since μ' is feasible, assigning i^{*t} to $\mu_{i^*}^{(t)'}$ while keeping all higher-priority assignments fixed admits a feasible completion. It implies $\mu_{i^*}^{(t)'} \in S^{\mu^{k-1}}_{f-av}(i^{*t})$. Since $\mu_{i^*}^{(t)'}\ \pi_{i^*}\ \mu_{i^*}^{(t)}$ and $\mu_{i^*}^{(t)'} \in S^{\mu^{k-1}}_{f-av}(i^{*t})$, the slot selected by the mechanism for replica i^{*t} cannot be the π_{i^*}-maximal element of $S^{\mu^{k-1}}_{f-av}(i^{*t})$. Thus, no such μ' exists, and μ is Pareto efficient. ■

Appendix B Capacity–Cluster Balance Beyond Quota Two

Theorem 1 identifies two distinct sources of DA's uniform feasibility guarantee under quota two. The first is capacity–cluster balance: when the number of clusters equals the slot quota, every full feasible slot contains one agent from each cluster. The second is the nonbinding case in which every cluster is a singleton. This appendix extends the characterization to a homogeneous slot quota $q \geq 2$.

Consider a single-unit-demand environment with $K = |S|$ slots, each with quota q, so that $|A| = qK$. Let $A_1, \ldots, A_m$ be the proximity clusters. We use the natural quota-q extension of the priority-respecting choice rule: a slot considers applicants in the common priority order and retains an agent if her cluster is not already represented, until either q agents have been retained or no additional cluster is represented. For each slot, this rule is precisely the fixed-priority greedy choice induced by the rank-q laminar matroid defined in Section 3.1.2. It is therefore substitutable for each $q \geq 2$ (Yokoi, 2019).

We use the following feasibility observation. If a population with cluster sizes $b_1, \ldots, b_r$ satisfies $\sum_{\ell=1}^{r} b_\ell = qL$ and $b_\ell \leq L$ for every ℓ, then it can be assigned to L quota-q slots so that each slot contains agents from q distinct clusters. This follows from an integral-flow construction with capacity b_ℓ from the source to cluster ℓ, unit capacity from each cluster to each slot, and capacity q from each slot to the sink. The relevant Hall-type inequalities hold because, for every set R of clusters, $\sum_{\ell \in R} b_\ell \leq L \min\{|R|, q\}$.

Proposition 3 *Fix a single-unit-demand environment with homogeneous slot quota $q \geq 2$ and m proximity clusters.*

1. *If $m < q$, no complete feasible allocation exists.*

2. *Suppose that a complete feasible allocation exists. DA produces a complete feasible allocation for every common priority ordering and every preference profile if and only if either $m = q$ or every proximity cluster is a singleton.*

3. *Consequently, if the zone-coverage constraint is non-vacuous, DA has a uniform complete-feasibility guarantee if and only if $m = q$.*

Proof. If $m < q$, a slot can contain at most one agent from each of the m clusters and therefore cannot be filled to quota. Hence no complete feasible allocation exists.

Suppose that $m = q$ and that a complete feasible allocation exists. Every full feasible slot must contain one agent from each cluster. Since there are K slots, complete feasibility implies $|A_\ell| = K$ for every ℓ. Each slot therefore provides one cluster-specific position for every cluster. Under the priority-respecting choice rule, an acceptance from one cluster does not reduce the position available to another cluster. Restricted to any cluster, DA is the standard one-to-one deferred-acceptance procedure between the K agents in that cluster and the K slots. It assigns every agent, so every slot receives one agent from each of the q clusters and the outcome is complete and feasible.

Suppose instead that every cluster is a singleton. Then every pair of distinct agents is compatible, and DA reduces to the standard quota-q procedure. If an agent were unmatched, she would have been

rejected by every slot, implying that every slot holds q agents. The slots would then hold all $qK = |A|$ agents, contradicting the existence of an additional unmatched agent. Hence DA produces a complete feasible allocation.

For the converse, suppose that $m > q$ and that at least one cluster is non-singleton. Let C be a largest cluster, with $|C| \geq 2$. Complete feasibility implies $|A_\ell| \leq K$ for every ℓ. Let H be the set of clusters other than C that contain exactly K agents. If $H \neq \emptyset$, maximality of C implies $|C| = K$. Hence $|H| \leq q - 2$; otherwise, C together with at least $q - 1$ other clusters of size K would exhaust the total population qK, leaving no agents for the additional clusters required by $m > q$. Because $m > q$ and $|H| \leq q - 2$, we can choose $q - 2$ distinct clusters other than C that include every cluster in H.

Select distinct agents $c, u \in C$ and one agent from each of the selected $q - 2$ clusters. After removing these q agents, the residual population has size $q(K - 1)$, and every residual cluster has size at most $K - 1$. By the feasibility observation above, the residual agents can be assigned feasibly to $K - 1$ slots, each filled with q agents from distinct clusters.

Let s_0 be the remaining slot. Every residual agent ranks her assigned slot first. Agents c, u, and the selected agents from the other $q - 2$ clusters rank s_0 first. Complete all remaining preferences arbitrarily, with u ranking the other slots after s_0, and choose a common priority ordering in which u has the lowest priority and $c \succ u$.

In the first round, each of the $K - 1$ residual slots receives and retains its designated feasible set of q agents. Slot s_0 receives q proposals from only $q - 1$ distinct clusters: c and u belong to C, while the other $q - 2$ applicants belong to distinct clusters. It retains c and the other $q - 2$ agents and rejects u. Agent u is subsequently rejected by every remaining slot, each of which is already filled with q higher-priority agents from distinct clusters. DA therefore terminates with u unmatched and one vacant position at s_0, although a complete feasible allocation exists.

Thus DA has a uniform complete-feasibility guarantee exactly when $m = q$ or the spatial constraint is nonbinding because every cluster is a singleton. ■

Proposition 3 shows that the number two in Theorem 1 is not intrinsically special. With quota two, $m = 2$ is the balanced case, while three clusters are the first non-vacuous case in which the number

of clusters exceeds the slot quota. When $m > q$ and the constraint is non-vacuous, a full slot can be formed from a strict subset of clusters, so locally admissible acceptances may alter residual cluster demands and eliminate all feasible completions. DA does not test for this residual effect; FSD and GFSD do so by requiring each tentative assignment to preserve completion feasibility.

Appendix C A Matroid-Union of Residual Completion: A Characterization Result

This appendix provides an alternative matroidal characterization of the residual-completion condition in Theorem 4. Although the feasibility constraint at each individual slot is laminar and induces a substitutable choice function, completing a partial allocation requires the unassigned replicas to be distributed jointly across the residual slot-specific matroids.

Fix a partial allocation μ. Let

$$U^{\mu} = \bigcup_{\ell=1}^{m} A_{\ell}^{\Gamma}(\mu)$$

denote the set of unassigned replicas. For each slot $s \in \mathcal{S}$, let $q_s^{\mu} = q_s - |\mu(s)|$ denote its residual capacity, and let L_s^{μ} denote the set of clusters not represented at s under μ. Thus, replicas from cluster ℓ remain admissible at s if and only if $\ell \in L_s^{\mu}$.

For each slot $s \in \mathcal{S}$, define $M_s^{\mu} = \left(U^{\mu}, \mathcal{I}_s^{\mu}\right)$, where

$$\mathcal{I}_s^{\mu} = \left\{ I \subseteq U^{\mu} : \begin{array}{ll} |I| \le q_s^{\mu}, & \\ |I \cap A_{\ell}^{\Gamma}(\mu)| \le 1 & \text{for each } \ell \in L_s^{\mu}, \\ I \cap A_{\ell}^{\Gamma}(\mu) = \emptyset & \text{for each } \ell \notin L_s^{\mu} \end{array} \right\}.$$

Thus, $\mathcal{I}_s^{\mu}$ is the family of residual replica sets that can be added jointly to slot s: such a set cannot exceed the residual capacity of s, cannot contain two replicas from the same cluster, and cannot contain a replica from a cluster already represented at s.

Lemma 3 *For each partial allocation μ and slot $s \in \mathcal{S}$, the set system M_s^{μ} is a laminar matroid.*

Proof. Consider the family consisting of U^μ and the nonempty residual cluster sets $\{A_\ell^\Gamma(\mu) : A_\ell^\Gamma(\mu) \neq \emptyset\}$. This family is laminar since the residual cluster sets are pairwise disjoint and each is contained in U^μ. Assign capacity q_s^μ to U^μ. For each nonempty residual cluster set $A_\ell^\Gamma(\mu)$, assign capacity one when $\ell \in L_s^\mu$ and capacity zero when $\ell \notin L_s^\mu$. The subsets of U^μ satisfying these laminar capacity constraints are exactly the sets in $\mathcal{I}_s^\mu$. Hence, by the definition of a laminar matroid, M_s^μ is a laminar matroid. ■

A feasible completion of μ is equivalent to for each $s \in \mathcal{S}$, the existence of set $I_s \in \mathcal{I}_s^\mu$ such that $U^\mu = \bigcup_{s\in\mathcal{S}} I_s$. Indeed, I_s is the set of residual replicas assigned to slot s in the completion. Since total demand equals total slot capacity, $|U^\mu| = \sum_{s\in\mathcal{S}} q_s^\mu$. Moreover, for each $s \in \mathcal{S}$, $|I_s| \leq q_s^\mu$. Because the sets I_s partition U^μ, these inequalities jointly imply for each $s \in \mathcal{S}$, $|I_s| = q_s^\mu$. Thus, such a partition assigns each residual replica exactly once and fills every residual slot capacity while respecting the cluster restrictions.

To determine when such a partition exists, we apply Edmonds's matroid-union theorem, equivalently the matroid-partition theorem (Edmonds, 1968). Since this theorem is stated in terms of the rank functions of the slot-specific matroids, we first record the rank function of each M_s^μ. For each $X \subseteq U^\mu$, the rank function of M_s^μ is

$$r_s^\mu(X) = \min\left\{q_s^\mu, \left|\left\{\ell \in L_s^\mu : X \cap A_\ell^\Gamma(\mu) \neq \emptyset\right\}\right|\right\}. \tag{3}$$

Indeed, an independent subset of X can contain at most one replica from each cluster that is both represented in X and available at s, and it can contain at most q_s^μ replicas in total. Conversely, selecting one replica from each such cluster, up to q_s^μ replicas, attains the displayed bound.

Edmonds's matroid-union theorem now implies that a partition of U^μ into sets $I_s \in \mathcal{I}_s^\mu$, one for each $s \in \mathcal{S}$, exists if and only if for each $X \subseteq U^\mu$, $|X| \leq \sum_{s\in\mathcal{S}} r_s^\mu(X)$. The following proposition shows that these rank inequalities are exactly the cluster-based completion inequalities in Theorem 4.

Proposition 4 *For a fixed partial allocation μ, the following conditions are equivalent:*

(i) *For each $X \subseteq U^\mu$,*

$$|X| \leq \sum_{s\in\mathcal{S}} r_s^\mu(X).$$

(ii) *For each* $L \subseteq \{1, \dots, m\}$,

$$\sum_{\ell \in L} |A_\ell^\Gamma(\mu)| \le \sum_{s \in \mathcal{S}} \min\left\{|L_s^\mu \cap L|, q_s^\mu\right\}.$$

Proof. Suppose first that condition *(i)* holds. Fix $L \subseteq \{1, \dots, m\}$, and let $L^+ = \left\{\ell \in L : A_\ell^\Gamma(\mu) \neq \emptyset\right\}$. Removing clusters with zero residual demand leaves the left-hand side of condition *(ii)* unchanged and weakly decreases its right-hand side. It is therefore sufficient to establish the inequality for L^+.

Set $X = \bigcup_{\ell \in L^+} A_\ell^\Gamma(\mu)$. Because the residual cluster sets are pairwise disjoint, $|X| = \sum_{\ell \in L^+} |A_\ell^\Gamma(\mu)|$. Moreover, for each slot $s \in \mathcal{S}$, the clusters represented in X and available at s are exactly those in $L_s^\mu \cap L^+$. Hence, by (3), $r_s^\mu(X) = \min\left\{|L_s^\mu \cap L^+|, q_s^\mu\right\}$. Applying condition *(i)* to X gives

$$\sum_{\ell \in L^+} |A_\ell^\Gamma(\mu)| \le \sum_{s \in \mathcal{S}} \min\left\{|L_s^\mu \cap L^+|, q_s^\mu\right\}.$$

The same inequality therefore holds for L, so condition *(ii)* follows.

Conversely, suppose that condition *(ii)* holds. Fix $X \subseteq U^\mu$, and let $L(X) = \left\{\ell \in \{1, \dots, m\} : X \cap A_\ell^\Gamma(\mu) \neq \emptyset\right\}$. Since X contains only replicas from clusters in $L(X)$, $|X| \le \sum_{\ell \in L(X)} |A_\ell^\Gamma(\mu)|$. Applying condition *(ii)* to $L(X)$ gives

$$|X| \le \sum_{s \in \mathcal{S}} \min\left\{|L_s^\mu \cap L(X)|, q_s^\mu\right\}.$$

By the definition of $L(X)$ and (3), for each $s \in \mathcal{S}$, $\min\left\{|L_s^\mu \cap L(X)|, q_s^\mu\right\} = r_s^\mu(X)$. Consequently,

$$|X| \le \sum_{s \in \mathcal{S}} r_s^\mu(X).$$

■

Relationship to Theorem 4. Condition *(ii)* in Proposition 4 is exactly the completion condition in Theorem 4. Together with Edmonds's theorem, the proposition therefore establishes that residual completion is a matroid-union problem: the unassigned replicas must be partitioned into sets that are independent in the respective residual slot matroids.

This characterization also clarifies the distinction between local substitutability and global completion. Each slot's fixed-priority greedy choice is substitutable because it operates under a laminar matroid. Nevertheless, a tentative assignment changes the common ground set of unassigned replicas and the associated residual slot matroids. At the receiving slot, it also reduces residual capacity and removes the assigned replica's cluster from the set of available clusters. Independently applied local choices therefore need not preserve the Edmonds rank inequalities required for a complete extension.

Relationship to the maximum-flow formulation. The matroid-union and maximum-flow formulations provide equivalent characterizations of the same residual-completion problem. The matroid-union formulation places the problem within the established matroidal-constraint framework. The maximum-flow formulation exploits the additional structure of the cluster domain to provide a direct constructive and computational test: a single maximum-flow computation determines whether a partial allocation admits a completion, and an integral maximum flow directly identifies one. This residual maximum-flow test is the completion-feasibility oracle used at each step of $GFSD$.

Appendix D No Justified-Envy under Lexicographic Preferences

The no justified-envy property used in Theorem 5 is the direct application of Definition 3 to the unit-demand replica market. It evaluates priority claims according to the replica order $\succ^{\Gamma}$ and the slot preferences inherited by the replicas. An alternative is to formulate no justified-envy directly by comparing agents' complete bundles under the lexicographic extension.

Definition 8 *An allocation μ satisfies **strong no justified-envy** if for each $(i,j)_{\succ}$ with $\mu(j)\ \pi_i^{\text{lex}}\ \mu(i)$ and each feasible allocation μ' with $\mu'(i)=\mu(j)$, there exists $k \succ i$ such that $\mu(k)\ \pi_k^{\text{lex}}\ \mu'(k)$.*

The $GFSD$ outcome need not satisfy Definition 8. The failure arises because the layered replica order may place an early replica of a lower-priority agent ahead of a later replica of a higher-priority agent.

Example 3 *Let $\mathcal{A}=\{i,j,k\}$ and $\mathcal{S}=\{s_1,s_2,s_3\}$. Each agent demands two slots, and every slot has quota two. That is, $d_i=d_j=d_k=2$ and $q_{s_1}=q_{s_2}=q_{s_3}=2$.*

Let each agent form a singleton proximity cluster, so each pair of distinct agents is zone-covering. The agent priority order is $i \succ j \succ k$, and the agents' preferences are as follows: $s_1 \; \pi_i^P \; s_2 \; \pi_i^P \; s_3$, $s_2 \; \pi_j^P \; s_1 \; \pi_j^P \; s_3$, and $s_2 \; \pi_k^P \; s_1 \; \pi_k^P \; s_3$. The layered replica order is $i^1 \succ^\Gamma j^1 \succ^\Gamma k^1 \succ^\Gamma i^2 \succ^\Gamma j^2 \succ^\Gamma k^2$.

Under GFSD, the replicas are assigned as follows:

$$\mu(i^1) = s_1, \qquad \mu(j^1) = s_2, \qquad \mu(k^1) = s_2,$$

$$\mu(i^2) = s_3, \qquad \mu(j^2) = s_1, \qquad \mu(k^2) = s_3.$$

At every step, the indicated assignment is the most-preferred feasibly available slot; the remaining assignments displayed above provide a feasible completion of the corresponding partial allocation. The induced allocation is therefore

$$\mu(i) = \{s_1, s_3\}, \qquad \mu(j) = \{s_1, s_2\}, \qquad \mu(k) = \{s_2, s_3\}.$$

Under agent i's lexicographic preference, $\mu(j) \; \pi_i^{\text{lex}} \; \mu(i)$, since the two bundles have the same best slot s_1, while i prefers s_2 to s_3. Consider the alternative allocation

$$\widehat{\mu}(i) = \{s_1, s_2\}, \qquad \widehat{\mu}(j) = \{s_1, s_3\}, \qquad \widehat{\mu}(k) = \{s_2, s_3\}.$$

This allocation is feasible: every agent receives two slots, every slot receives two agents, and every slot contains agents from distinct proximity clusters. Moreover, $\widehat{\mu}(i) = \mu(j)$. Since i is the highest-priority agent, there exists no agent $h \succ i$ such that $\mu(h) \; \pi_h^{\text{lex}} \; \widehat{\mu}(h)$. Hence, the GFSD outcome violates strong no justified-envy.

Example 3 does not contradict the fairness result in Theorem 5. The slot s_2 desired by i at the second position of her bundle is occupied by j^1 and k^1, both of whom precede i^2 under $\succ^\Gamma$ and rank s_2 first. In particular, the alternative allocation $\widehat{\mu}$ moves j^1 away from s_2 and therefore does not preserve the assignments of all replicas preceding i^2. The example thus separates priority protection for marginal units under $\succ^\Gamma$ from priority protection for complete bundles under $\succ$.

Appendix E The value of $GFSD$: Further Evidence

We also examine how preference correlation interacts with congestion (Table 3). For each ρ, preferences are generated by combining official duty shift coefficients with idiosyncratic noise (explained in detail in Section F.2 below). *Peak pressure*—the maximum ratio of top-three demand to duty shift capacity—captures the severity of feasibility bottlenecks. Across all calibrations, MSD occasionally produces assignment failures, which vary across preference environments. It reflects violations of the completion condition in Theorem 4. In contrast, $GFSD$ always clears the market, demonstrating robustness to demand concentration.

The more severe failure occurs under Fixed-Pair $GFSD$. Even when the completion-preserving logic of $GFSD$ is retained, restricting availability to the SQ's observed partner graph leaves roughly 205–209 replicas unassigned on average. This shows that the value of $GFSD$ comes not only from the look-ahead feasibility check, but also from the structural flexibility created by allowing admissible partnerships beyond historically observed pairs.

Table 3: Preference Alignment, Congestion, and Assignment Failures

Preference Scenario	**Peak Pressure (Top-3)**	MSD **Unassigned**	$GFSD$ **Unassigned**	**Fixed-Pair** $GFSD$ **Unassigned**
$\rho = 0.0$	18.0	3.00	0.00	205.38
$\rho = 0.3$	42.0	1.62	0.00	207.38
$\rho = 0.5$	98.0	1.75	0.00	208.50
$\rho = 0.7$	170.8	1.12	0.00	206.38
$\rho = 0.9$	282.8	1.38	0.00	205.88

Notes: For each ρ, preferences are generated from official duty shift coefficients and an idiosyncratic noise term. Peak pressure is the maximum, across duty shifts, of the ratio between aggregate top-three demand and duty shift capacity, averaged across the cluster-recovery and distance-specification scenarios. Unassigned counts refer to unmatched replicas.

Figure 3 visualizes the structural channel behind these results in the Ankara/Keçiören district-level subsample. Panel A (SQ, persistent partnerships) shows 21 disconnected components, creating topo-

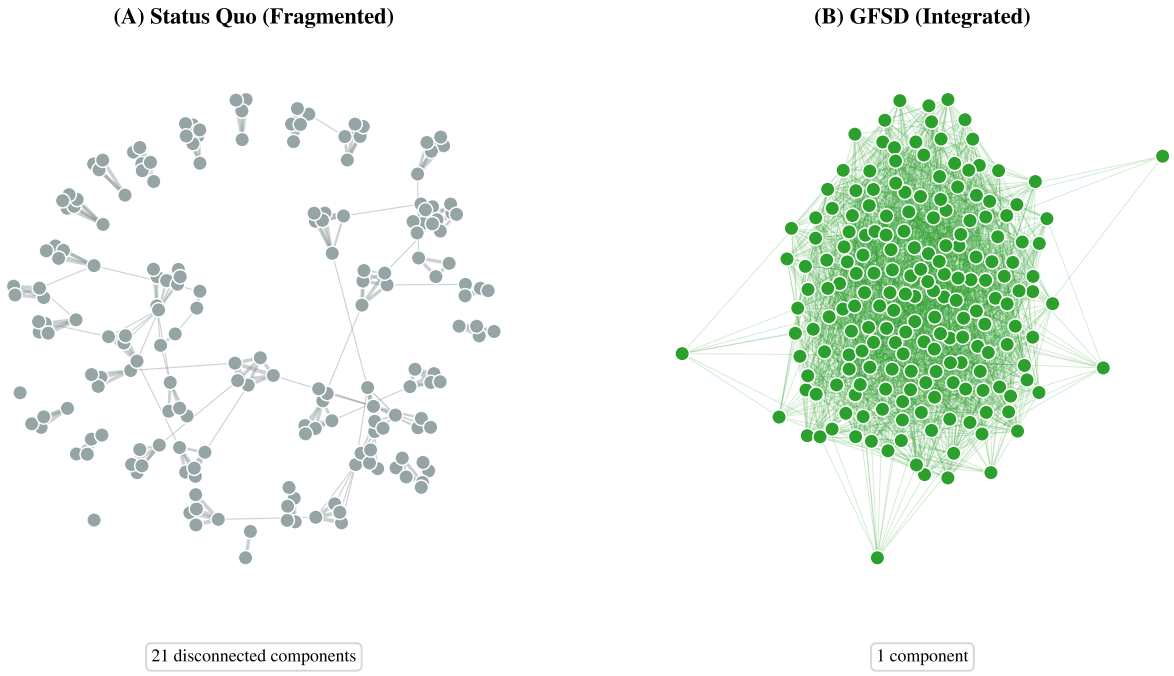


Figure 3: **Structural transformation of the market.**
The figure uses the Ankara/Keçiören district-level subsample and depicts the 188 pharmacies with positive 2024 duty demand. Nodes represent active pharmacies; edges represent realized co-assignments.

logical bottlenecks. Panel B ($GFSD$, realized co-assignments) forms a single component spanning all 188 active pharmacies, in which the realized co-assignment network expands from a sparse collection of disconnected components (high modularity) into a connected structure (near-zero modularity). This integration eliminates deadlocks, and improves welfare (Figure 4). SQ resembles assignment largely insensitive to preference heterogeneity (median rank 180). Fixed-Pair $GFSD$ shows modest improvement but remains constrained by partner rigidity (median rank 61.9). Under the simulated preference profiles, $GFSD$ shifts the assigned-rank distribution toward better-ranked duty shifts relative to the status quo and Fixed-Pair $GFSD$.

E.1 Robustness and Simulation Variability

Table 4 summarizes the robustness exercises that vary the recovered cluster structure and the distance specification used in the recovery stage. The qualitative ranking of mechanisms is unchanged. $GFSD$ clears the market in every specification. MSD remains close to $GFSD$ in many cases, but it lacks a uniform clearing guarantee and leaves up to four replicas unassigned under alternative cluster re-

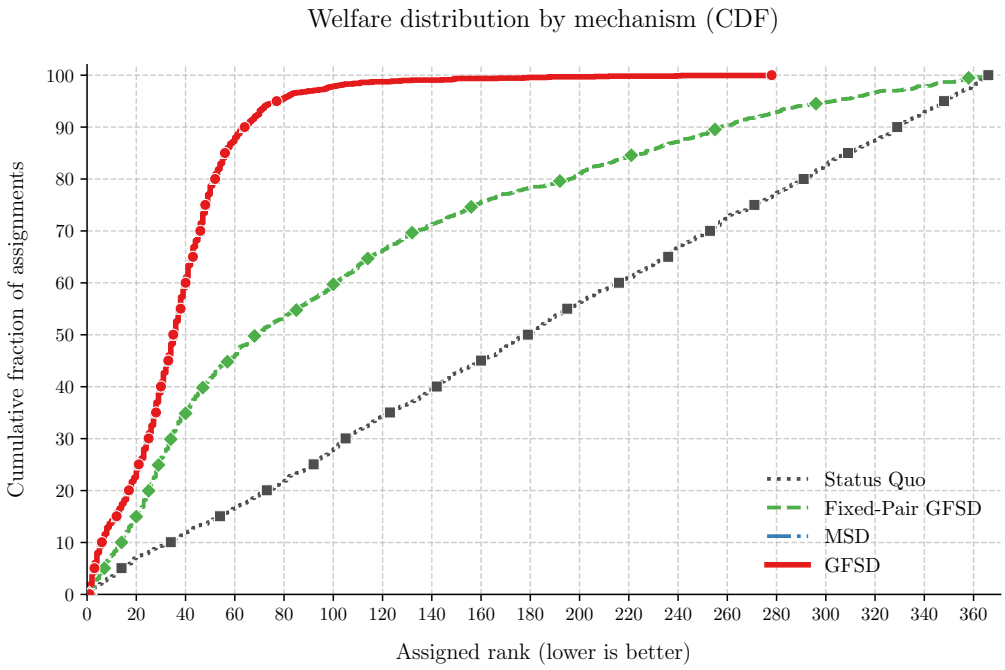


Figure 4: **Model-Implied Assigned-Rank Distributions under Simulated Preferences.**

The figure reports cumulative distributions of assigned ranks under the SQ, Fixed-Pair $GFSD$, MSD, and $GFSD$. Lower ranks indicate better preference satisfaction. Rank distributions for mechanisms with unassigned replicas are conditional on assigned replicas.

constructions. Fixed-Pair $GFSD$ is consistently constrained by SQ's observed partner graph, leaving roughly 195–219 replicas unassigned across these robustness exercises.

Table 4: Robustness across Cluster and Distance Specifications

Robustness dimension	**Specifications**	$GFSD$ **Unassigned**	MSD **Unassigned**	**Fixed-Pair** $GFSD$ **Unassigned**
Cluster reconstruction	Repeated co-duty thresholds $t \in \{2, 3, 5, 7, 10\}$, with recovered $K^\star \in \{10, 11, 12\}$	0–0	0–4	195–216
Distance specification	Linear, unpenalized squared, and Haversine distance, holding $t = 3$, $K^\star = 11$, and $\rho = 0.7$ fixed	0–0	0–2	196–219

Because pharmacy preferences are not elicited in the administrative data, the simulations are counterfactual exercises under reconstructed environments rather than estimates based on observed preference data. We therefore summarize sensitivity to the maintained reconstruction and calibration choices. Table 5 reports scenario-level standard deviations and ranges for the principal feasibility and rank-based welfare outcomes across the 40-scenario grid. These statistics describe variation across the specifications considered; they are not confidence intervals or sampling standard errors.

Table 5: Variation across Robustness Scenarios

Mechanism	Unassigned			Mean Rank			Top-3 (%)		
	Mean	SD	Range	Mean	SD	Range	Mean	SD	Range
Status Quo	0.00	0.00	0–0	181.4	2.9	179.3–186.9	1.2	0.2	0.8–1.3
Fixed-Pair GFSD	206.70	7.18	194–220	100.9	12.5	86.7–124.3	9.1	7.7	1.0–21.4
MSD	1.78	1.89	0–8	29.0	20.4	8.9–64.8	15.7	11.4	1.9–32.9
GFSD	0.00	0.00	0–0	29.3	20.4	9.2–65.4	15.7	11.4	1.9–32.8

Notes: Entries summarize variation across the full robustness grid of 40 scenarios. Ranges report the minimum and maximum scenario-level values. Rank-based welfare statistics are computed conditional on assigned replicas; unassigned counts are reported separately. Lower rank indicates better preference satisfaction.

The qualitative ranking of mechanisms is stable across the grid. $GFSD$ clears the market in every scenario, with zero variation in unassigned replicas. MSD delivers nearly identical assigned-rank outcomes, but leaves up to eight replicas unassigned in some scenarios. Fixed-Pair $GFSD$ remains substantially constrained by the status quo's observed partner graph, leaving between 194 and 220 replicas unassigned across scenarios. Thus, the model-implied welfare gains of flexible assignment are robust, but only $GFSD$ combines these gains with uniform clearing.

Appendix F Data Generation

F.1 Zone-Coverage Relation

While geographical coordinates are available, Euclidean distance alone fails to induce a negatively transitive binary relation, consequently, a partition. Data reveals only a subset of zone-covering partners that are frequently matched together. This leaves the relation between the unobserved partners ambiguous.

We adopt a constructive approach to recover the zone-coverage. For a repeated co-duty threshold t, let E_t denote the graph whose edges are pharmacy pairs that were historically assigned to the same duty shift at least t times. Each edge $(i, j) \in E_t$ is treated as hard evidence that pharmacies i and j

belong to different service zones. Equivalently, for each $(i, j) \in E_t$, the recovered partition C_t must satisfy $C_t(i) \neq C_t(j)$. Among partitions satisfying these hard cross-zone constraints and the cluster-size bounds, we select geographically compact partitions.

The recovery proceeds in two steps. First, for each threshold t, we compute the smallest certified feasible number of clusters $K_t^{\min}$ satisfying the hard co-duty constraints and the size bounds. Second, beginning from $K_t^{\min}$, we solve a compactness grid over larger values of K and select the smallest $K_t^\star$ after which geographic coherence stabilizes. We repeat the recovery for $t \in \{2, 5, 7, 10\}$. These alternatives are visualized in Figure 5.

We also assess sensitivity to the geographic criterion used in the recovery procedure. Holding the co-duty threshold and the number of clusters fixed at their baseline values, $t = 3$ and $K^\star = 11$, we compare the baseline squared-distance specification with a far-distance penalty against three alternatives: a linear-distance specification, a squared-distance specification without the far-distance penalty, and a squared Haversine-distance specification with the far-distance penalty. These exercises isolate sensitivity to the functional form of the geographic compactness criterion while holding the co-duty constraints and the number of recovered clusters fixed.

The induced relation $\mathcal{F}$ contains exactly the distinct cross-cluster pairs and is therefore symmetric, irreflexive, and negatively transitive. Indeed, the complement of $\mathcal{F}$ is the equivalence relation of belonging to the same recovered service zone. Hence, the recovered relation is both theoretically consistent with the partition structure required by the mechanism and empirically grounded in the district's geography and observed co-duty data.

Figure 5 provides an illustrative visualization of the recovered service zones in the Ankara/Keçiören district-level subsample under alternative repeated co-duty thresholds. The clusters exhibit clear geographical coherence across specifications. Panel B corresponds to the baseline partition, $t = 3$ and $K^\star = 11$, used in the main mechanism simulations. The remaining panels show the robustness partitions for $t \in \{2, 3, 5, 7, 10\}$. Colored regions correspond to convex hulls of the recovered clusters and are included only as visual aids; they are not imposed constraints in the recovery procedure. Pharmacies within the same recovered service zone are geographically concentrated, and the broad spatial

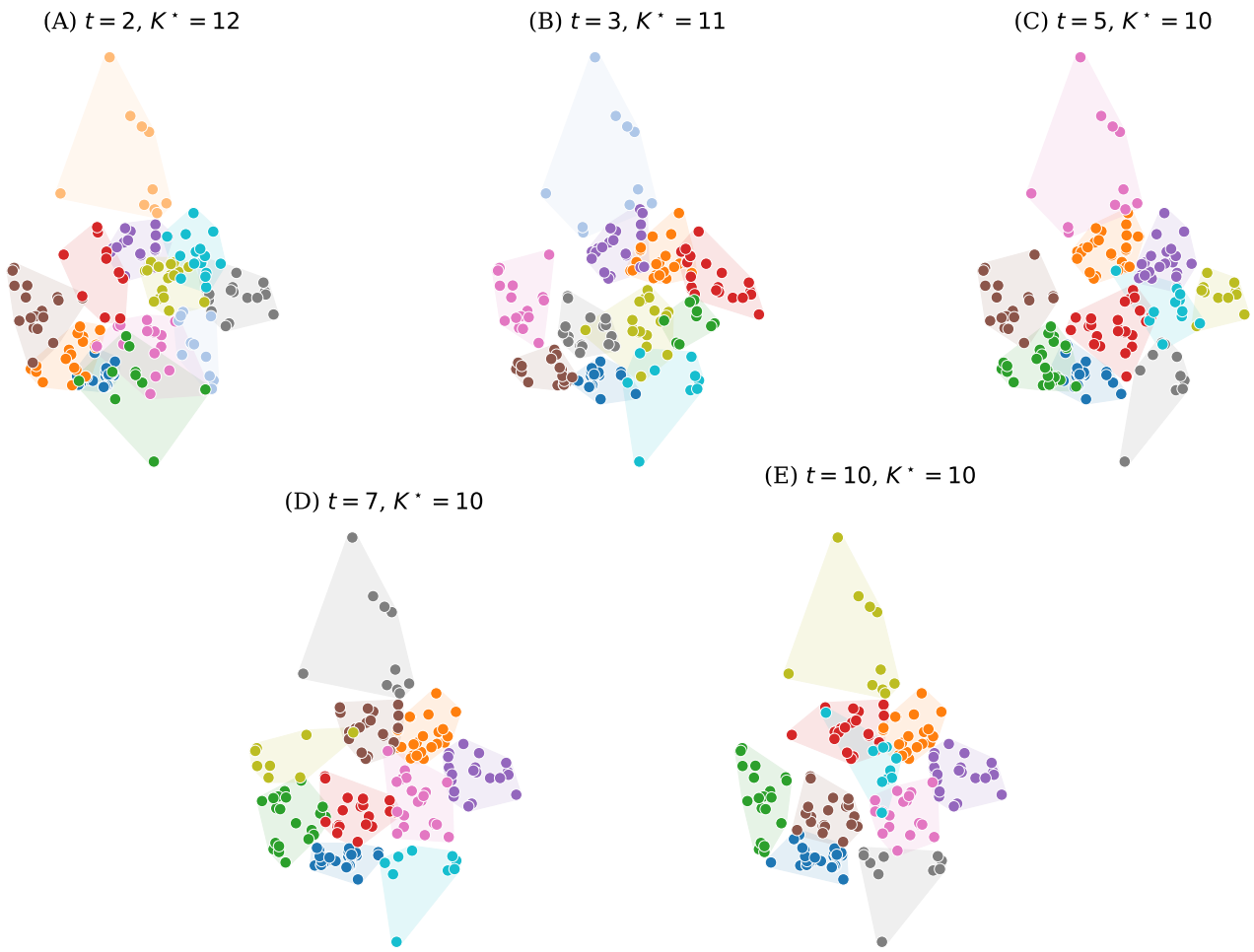


Figure 5: **Recovered service zones.**
The figure depicts the full 198-pharmacy cluster-recovery sample from the Ankara/Keçiören district-level subsample.

structure of the partition remains stable as the hard co-duty threshold varies.

F.2 Preferences and Priority Ordering

We construct a Data-Generating Process (DGP) that produces synthetic but behaviorally disciplined preference profiles. The financial attractiveness of the slot and an idiosyncratic component derives the utility of a pharmacy for a specific slot. Formally, the utility of pharmacy i for a slot s, denoted as $u_{i,s}$, is as follows:

$$u_{i,s} = \rho \cdot Z_s + \sqrt{1 - \rho^2} \cdot \epsilon_{i,s}$$

Z_s represents the financial attractiveness of slot s, common to all pharmacies. We proxy Z_s using

the slot coefficients provided by a Region Representative of the Istanbul Chamber of Pharmacists. The coefficients summarize expected patient flow and workload, and thus correlate with the revenue potential of a slot. We normalize Z_s to have mean zero and unit variance.

We capture unobserved pharmacy-specific shocks that affect desirability of a slot by $\epsilon_{i,s} \sim \mathcal{N}(0, 1)$. These shocks are assumed independent identically distributed across i and s. The scalar $\rho \in [0, 1]$ adjusts the extent of preference alignment in the market. When ρ is high, preferences are largely determined by the revenue gradient embedded in Z_s. When ρ is low, pharmacies' preferences are driven mostly by idiosyncratic factors. The ordinal preference relation π_i for each pharmacy is derived by ranking the slots according to utilities such that for each $i \in \mathcal{A}$ and $s, s' \in \mathcal{S}$, $s \; \pi_i \; s'$ if and only if $u_{i,s} > u_{i,s'}$.

To ensure that our synthetic data accurately reflects the targeted market structure, we employ a Monte Carlo calibration procedure. For a given alignment parameter (e.g., $\rho = 0.7$), we generate $N = 5000$ realizations of the preference profile. For each realization, we compute the average Spearman's rank correlation coefficient between the generated preference lists and the baseline common value vector Z. We select the specific realization that minimizes the distance between the realized average correlation and the intended alignment level for that scenario. This approach ensures that the simulated environment is not merely random but is structurally consistent with the economic incentives derived from the Istanbul Chamber of Pharmacists' data. Because the official coefficient vector contains repeated and discrete values, the realized rank correlations need not coincide exactly with the input loading parameter ρ. We therefore report realized average Spearman correlations when summarizing the preference environments.

The priority ordering $\succ$ is determined by slot coefficients. For each pharmacy, we compute the cumulative coefficient over 2015–2023 and rank pharmacies from lowest to highest. This ranking serves as the empirical analogue of the exogenous priority order in the theoretical model and is consistent with institutional rotation: pharmacies historically assigned to lower-valued slots receive strictly higher priority.

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